Improving Student Learning through Adaptive Learning Platforms: Experimental Evidence from the Dominican Republic

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Abstract

This paper examines the impact of an intervention that combines a computer-adaptive learning (CAL) platform with group tutoring to accelerate learning recovery among high school students in the Dominican Republic. Using a randomized controlled trial, we introduced the CAL platform ALEKS, replacing a portion of regular math instruction for 9th graders and offering group tutoring to high-need students. Our findings indicate that the adaptive platform improved student math performance by 0.29 to 0.31 standard deviations, with robust effects across multiple knowledge domains. However, the addition of tutoring did not yield further improvements, potentially due to reduced platform engagement and implementation challenges. This study contributes to the literature on remedial education and AI-driven interventions, underscoring the potential of CAL platforms as scalable alternatives to traditional tutoring in resource-constrained settings.

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1 Introduction

In 2018, less than 20 percent of students in low and middle-income countries (LMIC) were able to meet minimum proficiency levels in mathematics (World Bank, 2017).[1] More recently, the learning crisis, already a pressing concern, has deepened as a result of the COVID-19 pandemic. Over the course of two years, from February 2020 to February 2022, schools around the world were closed for an average of seven months (UNESCO, 2021). To counter these negative impacts, various strategies have been explored, including in-person and online tutoring (Carlana and La Ferrara, 2021; Nickow et al., 2020), lesson delivery via phone calls (Angrist et al., 2020), and the use of Artificial Intelligence (AI) tools (Muralidharan et al., 2019), all of which have shown promising results in improving student outcomes. However, a pivotal question remains: how can school systems build more resilient methods of delivering instruction?

One promising approach lies in leveraging technology, particularly AI, to address the growing challenges associated with providing quality education at-scale. AI-powered tools offer the potential to accelerate learning recovery and transform education systems worldwide. Among these, computer-adaptive learning (CAL) platforms stand out as highly effective (Antoninis et al., 2023). In classrooms with diverse learning levels, where teachers may struggle to address individual needs, CAL tools adapt lessons based on each student's responses, tailoring activities to match their abilities—all at a fraction of the cost of one-on-one tutoring. Additionally, they provide students with instant feedback and equip teachers with valuable data to track progress and performance.

In this study, we evaluate through a multi-stage randomized controlled trial (RCT) the impact of an innovative intervention combining the use of a CAL platform with group tutoring, with the goal of accelerating learning recovery among high school students in the Dominican Republic (DR). While the introduction of these new technologies may accelerate learning for all students within a classroom, it may not suffice to significantly narrow the substantial learning disparities that often exist among students. With this goal in mind, we combine the use of CAL with targeted group tutoring for students most in need. The study design and survey instruments allow us to generate evidence on whether the use of CAL, by itself, or when combined with targeted tutoring, can yield improvements in student outcomes.

In 2023, our intervention introduced the use of CAL as part of regular instruction for 9th graders, replacing two out of the seven weekly hours allocated for regular math instruction. Randomization was conducted at the classroom level (39 classrooms in total). The software chosen, ALEKS, developed by McGraw Hill, has been adapted to the DR math curriculum and was used in approximately 60 schools across the country, yielding satisfactory qualitative results.[2] All students in treated classrooms received free licenses to access this software. Additionally, within the selected classrooms, we randomized access to in-person tutoring by providing up to three weekly hours of tutoring to high-need students. This component aimed to explore complementarities between CAL and more traditional forms of instruction. Students participating in tutoring sessions had access to the platform, and tutors were instructed to use the platform data to monitor student progress, identify learning gaps, and plan their lessons. These additional lessons were conducted during school hours otherwise dedicated to extracurricular activities.

We find that the use of the adaptive software during math classes improved student outcomes by

[1]This knowledge threshold indicates that students below this level have not yet acquired basic math skills, including simple calculations with whole numbers, or interpreting bar graphs.

[2]The intervention was carried out by the Ministry of Education from 2018 to 2023, with support from the World Bank.

0.29 to 0.31 standard deviations by endline (roughly six months after the start of the intervention). These positive effects are robust across various model specifications and consistent across multiple knowledge domains, including algebra and geometry, as well as different difficulty levels, such as content below and at grade level. However, we do not find statistically significant differences between classes that received the tutoring component and the control group, suggesting that adding tutoring to the CAL component may have hindered the effectiveness of the latter.

Why did the two treatment arms have different impacts? While we do not find conclusive evidence explaining these negative differences, we observe that students selected for tutoring were less likely to use the adaptive platform (as were their classmates), exhibited lower levels of locus of control (i.e., belief in their ability to influence their own outcomes), and experienced lower levels of math anxiety compared to the CAL-only group. Additionally, we encountered significant implementation challenges that may have affected the quality and consistency of the tutoring sessions, including high student absenteeism. We also find evidence indicating that the tutoring may not have been well-targeted, as students selected by teachers to receive tutoring were not among the lowest performers at baseline, as was intended by the program.

Our paper contributes to the growing literature on educational remediation strategies, particularly by exploring the effectiveness of combining multiple approaches to address learning gaps. Tutoring has been shown to be one of the most effective interventions for improving learning outcomes. A recent meta-analysis by Nickow et al. (2020) reported an average effect size of 0.37 standard deviations for tutoring interventions, with one-to-one tutoring showing the largest effects. However, the scalability of this approach is hindered by high costs and the challenge of recruiting qualified tutors (Kraft and Falken, 2020; Nickow et al., 2020; Kraft et al., 2024), and the longstanding issue of finding scalable alternatives that are as effective, known as the "2-sigma problem" (Bloom, 1984), remains a significant concern. Recent evidence, however, suggests that small-group tutoring (up to five students per tutor) can achieve similar results at lower costs (Augustine et al., 2016; Guryan et al., 2023; Kraft and Falken, 2020; Nickow et al., 2020; Kraft et al., 2024). Our study contributes to this literature by examining the combination of small-group tutoring with the use of adaptive learning technologies in school, which could enhance the effectiveness of traditional group tutoring by leveraging CAL data to better address specific learning deficiencies.

Despite limited research on AI-driven education interventions (Miao et al., 2021), increasing access to computer-adaptive technologies has spurred a growing body of research (Angel-Urdinola et al., 2023; Guryan et al., 2023; Muralidharan et al., 2019).[3] Most existing studies have evaluated CAL in after-school settings or combined with tutoring (Muralidharan et al., 2019; Guryan et al., 2023), leaving open questions about its effectiveness when integrated into regular school schedules, or implemented alone (i.e. without a tutoring component) (Muralidharan et al., 2019). Additionally, because of the setting in which these studies have taken place, the impact of CAL alone versus additional instructional time remains difficult to disentangle (Glewwe and Muralidharan, 2016; Muralidharan et al., 2019). To our knowledge, this study is among the first to test the substitution of regular math instruction with the use of an adaptive learning platform (Muralidharan and Singh, 2023). We replace two out of seven weekly math instruction hours with CAL sessions, offering insight into its efficacy as part of the standard school curriculum.

[3] We focus on computer-adaptive learning, which personalizes content to students' actual learning levels, as opposed to traditional computer-assisted instruction, which reviews only grade-appropriate content. Studies of the latter include Angrist and Lavy (2002), Barrow et al. (2009), and Linden et al. (2003).

While most of the literature on teacher effectiveness has focused on measuring teacher value-added (Chetty et al., 2014a; Rockoff, 2004), identifying causal links between teacher effectiveness and student outcomes (Chetty et al., 2014b; Gershenson, 2016; Rivkin et al., 2005; Rockoff, 2004), or examining teacher characteristics (i.e. years of education or experience) associated with high value-added (Chingos and Peterson, 2011; Nye et al., 2004). It is less understood *how* to improve teacher effectiveness, and particularly, whether the introduction of new technology in the classroom can improve teacher effectiveness. Taylor (2015) evaluates the impact of computer-aided instruction on teacher effectiveness and finds that its introduction led to changes in how teachers' allocated class time. Using student-reported data, we explore whether the integration of an adaptive platform in class leads to changes in teachers' time use and practices, as this is one potential mechanism by which the intervention may have an impact on teacher effectiveness and student performance.[4] We do not observe changes in teachers' practices or allocation of time, as reported by students, suggesting that the effects observed may be attributed to the direct substitution of teachers' instruction for the use of the learning platform, rather than changes to how teachers allocated time and effort across tasks outside this period.

The deepening of the learning crisis due to COVID-19 highlights the need for more resilient education systems that can better withstand disruptions. Policymakers could invest in building infrastructure and capacity for blended learning models that integrate digital tools like CAL into teachers' work, ensuring continuity of learning in times of crisis. Given the high costs and logistical challenges associated with traditional tutoring, this study suggests that AI-driven adaptive learning platforms could serve as a scalable alternative for improving student outcomes. Expanding access to such technologies can provide affordable and effective learning support at scale, particularly in resource-constrained settings.

This paper proceeds as follows. In Section 2, we describe in detail the context, the intervention and its components. In Sections 3 and 4, we discuss the study design and describe our data. In Section 5, we present the empirical framework and the results of our analysis. Lastly, in Section 6, we conclude.

2 Intervention and Study Design

2.1 Background

The study takes place in the Dominican Republic (DR), a country facing substantial gaps in key knowledge areas such as mathematics and reading comprehension. In 2022, the DR ranked third to last in math on the OECD's Programme for International Student Assessment (PISA), with 92 percent of students classifying as low-performing (OECD, 2023a,b). The Dominican education system is divided into primary (grades 1-6), lower secondary (grades 7-9) and upper secondary (grades 10-12) levels, with students typically entering lower secondary around age 12. The school year in the DR is organized over two academic semesters, and usually begins in late August/early September and ends in late June.[5]

[4] It is important to note that the context of this study is different from that in Taylor (2015). While Taylor (2015) examines experiments that took place in the US, this work takes place in the Dominican Republic, one of the lowest performing countries in international assessments. For instance, when looking at the 2018 PISA Math results, the US ranked in the 9th place while the DR ranked 78th (out of 79 countries).

[5] The 2023-24 school year officially started on August 28th 2023 and ended on June 21st 2024.

The transition to secondary schooling represents a critical juncture in the academic trajectory of Dominican students. Previous research indicates that up to 40 percent of students drop out between 8th and 12th grade (IDEICE, 2017). These high dropout rates, coupled with notably low performance levels, underscore the urgency of implementing remedial interventions to improve student learning outcomes and retention. This challenge is not unique to the DR. Across Latin America and the Caribbean (LAC), countries face similar educational obstacles. The LAC region ranks in the bottom half of the global scale for educational quality in mathematics (IDB, 2023): the difference in performance between the average student in LAC and their OECD counterparts equates to approximately five years of learning.

Our study seeks to evaluate an intervention that provided targeted remedial training in mathematics to 9th grade students, addressing a key stage in their educational progression. At this point, students are likely to have accumulated significant learning gaps that could affect their chances of completing secondary education. By focusing on this age group, the intervention aims not only to enhance math proficiency but also to promote retention and grade progression. This work aims to contribute to broader efforts aimed at improving learning outcomes and reducing dropout rates in the Dominican Republic and across the wider LAC region.

2.2 The Program: APRENDE +

The intervention studied, APRENDE + ("Acompañamiento Personalizado para el Rendimiento Escolar" or "Personalized Support for School Performance"), was implemented between October 2023 and May of 2024 in a sample of 39 9th grade classrooms.[6] The program had two core elements: a CAL intervention and a tutoring component. The first component introduced the use of a CAL platform (ALEKS) for two hours per week as part of schools' regular schedule, substituting two out of seven weekly hours scheduled for math instruction. Half of the classrooms were randomly selected to receive CAL licenses for all students and their math teachers (stratified by classroom size). The second component introduced small-group tutoring during extracurricular hours for approximately 20 percent of students in randomly selected classrooms from the first treatment group. Each component is summarized in more detail below.

By testing the intervention in schools, as part of the regular school day, we examine its potential for scale and assess whether this type of approach can enhance regular math instruction. The study and intervention was funded by the Tinker Foundation and was implemented in partnership with the Ministry of Education of the Dominican Republic (MINERD) and the Abdul Latif Jameel Poverty Action Lab (J-PAL).[7]

ALEKS platform. The program provided access to ALEKS, a math-focused CAL software developed by McGraw Hill. ALEKS uses predictive algorithms to deliver personalized learning experiences through adaptive assessments and assignments. These tools evaluate students' knowledge by identifying the math topics they master, the ones they do not, and the ones they are ready to learn. This approach is grounded in Knowledge Space Theory (KST), which posits that a subject's knowledge can be mapped as a set of solvable problems (Doignon and Falmagne, 1985; Falmagne et al., 1990). Accordingly, ALEKS assesses students' mastery by their ability to solve representative problems from each topic.

[6]Figure 1 shows a full timeline of the study and intervention.

[7]The study is registered as AEARCTR-0012383 in the AER registry, and was approved under protocol number 24-444 by the Teachers College IRB.

The software encompasses over 600 math topics for 9th graders. Because it would not be feasible to cover all topics within an academic year, this list was narrowed to topics that were aligned to the targeted grade (9th grade). This included topics outlined in the 9th grade curriculum as well as others that were considered prerequisites for this grade. This process gave us a total of 337 topics which were organized into nine modules, each corresponding to different content areas scheduled for instruction during the academic year, as per curriculum guidelines.[8] In addition to the practice exercises included in each module, ALEKS offers functionalities that enable teachers to assign and evaluate homework and tests. It also provides a dashboard through which teachers can track the progress of their classes and individual students.

All teachers assigned to a CAL classroom received an initial 2-hour training on the use of the learning technology. This training was provided by McGraw Hill, the developer of the learning platform. In addition to this, during the first 3 months of the program, teachers met one-on-one with a CAL aide for approximately one hour every two weeks. The CAL aide's role was to support them in accessing and monitoring students' assessments, tracking their progress, and identifying topics that students had not yet mastered for lesson planning. Importantly, one of the primary aspects of this extended training was to help teachers utilize the software's specific tools to ease their workload—including features that allow assigning and automatically grading homework—thus enabling teachers to save time that could potentially be devoted to other teaching activities.

Tutoring Intervention. The tutoring component aimed to explore complementarities between CAL and more traditional forms of instruction, such as tutoring, by providing up to three weekly hours of tutoring to high-need students. Tutoring sessions were managed by one or two tutors, depending on the number of tutees per classroom, which typically ranged from 5 to 8. In each classroom selected to benefit from the tutoring component, we targeted approximately 20 percent of students (up to 8 students per classroom). At baseline, in order to select which students would receive tutoring, math teachers were instructed to select and rank up to 10 students for group tutoring based on need. To minimize bias in teachers' selection, after their first selection, we informed them of students' results in baseline math tests at the time of ranking. About 70 percent of teachers modified their selection after receiving this information.

Tutors were selected through a competitive process that included interviews conducted by the research team. All tutors were either recent graduates or in the final stages of earning a Bachelor's degree in Mathematics with a focus on secondary education, and had experience in using information technology for teaching. Tutoring sessions were integrated into students' schedules as elective workshops, covering approximately three academic hours per week, typically divided in one to three weekly sessions.[9] Elective workshops are offered for all students attending extended-day schools in the Dominican Republic. These types of schools offer school days that start at 8am and end at 4pm, and the workshops typically cover areas outside of the core curriculum, such as computing or art lessons.

To ensure quality and consistency in the tutoring sessions, each tutor was supported by a supervisor, who also had training and experience in mathematics education for high school students. The supervisor's role included reviewing individual session plans before each session, ensuring that the topics and instructional methods had been prepared before the session, and providing feedback

[8] This selection was performed by two local experts on math pedagogy based on a revision of the curriculum. The majority of topics included as prerequisites were identified by the learning software.

[9] Out of 9 tutoring classrooms, 2 sections had one weekly meeting of 3 hours, 5 sections had two weekly meetings of 2 and then 1 hour, and 2 sections had three meetings for 1 hour each day.

to tutors on their session plans if needed. In addition, tutors maintained close communication with the mathematics teacher to align their sessions with the classroom curriculum and were instructed to use the CAL platform for the tutoring sessions.[10] During the sessions, tutors reviewed topics covered in class during that week or revisited previous material, including pre-requisite topics that students found challenging. For instance, if a class was working on linear equations, a tutor may have reviewed the rules of signs before addressing the current topic.

The effectiveness of the tutoring sessions was monitored through unannounced visits by a member of the research team or the supervisor. These visits were intended to identify potential issues or areas for improvement. One-on-one meetings between the supervisor and the observed tutor were arranged after these visits to exchange feedback. Additionally, to monitor attendance, tutors were asked to complete a weekly survey to report the number of tutoring hours conducted for each class and student.

3 Implementation

3.1 Sample and Randomization

The intervention was administered in 9 public high schools of the Dominican Republic. Schools were selected based on whether they met the following conditions: (1) having an active computer lab, (2) having regular access to electricity and internet, (3) having an extended-day format,[11] and (4) having at least three 9th grade classrooms. In addition to this, we limited our sample to schools who had participated in *Programate*, a joint collaboration between MINERD and the World Bank that provided access to the ALEKS software to approximately 60 schools between 2018 and 2023. This set of conditions gave us an initial list of 34 schools. Between April and August 2023, the research team, with support of MINERD, contacted this initial list of schools via phone to confirm whether they met these conditions and to assess their interest in participating in this intervention. Participation in the intervention was voluntary and all contacted schools expressed strong interest in participating. However, out of this initial list of 34 schools, only 14 confirmed having access to internet and a computer lab.

Between September 4th and 15th, a member of the research team conducted school visits to validate the information collected during the Spring and Summer terms. Only 9 of the 14 visited schools had access to internet and functional computers. All participating schools were located in Santo Domingo and Santiago. Being the two largest cities in the country, they also have the largest concentration of students.[12]

Treatment was assigned using a two-stage randomization process at the classroom level. Initially, classrooms were ranked by size and paired with the closest matching classroom in terms of size. Within each pair, one classroom was randomly assigned to the treatment group and the other to the control group, ensuring similarly sized classrooms across groups. This pairing and

[10] For example, tutors were encouraged to use the platform to identify topics where students were lagging behind and review these during tutoring sessions, or use the platform's accompanying exercises as practice. If a given session had a higher tutee/tutor ratio due to a tutor absence, the tutor was encouraged to alternate the use of CAL with small-group tutoring.

[11] Approximately 67 percent of public high schools in the Dominican Republic offered an extended school day format during the 2021-2022 school year (MINERD, 2023), where the school day starts at 8am and ends at 4pm.

[12] According to MINERD's Educational Statistics Yearbook for 2021-2022, approximately 45 percent of students were located in Santo Domingo and Santiago.

randomization process was repeated within the treatment group to assign half of classrooms to the group tutoring component. The study ultimately comprised 38 classrooms, with 19 assigned to receive the CAL software (treatment group) and the remaining 19 serving as controls.[13] Of the classrooms in the treatment group, 9 were further selected to receive additional group tutoring. The randomization design is summarized in Figure 2 and power calculations are included in Appendix A.2.1.

3.2 Implementation: CAL Component

The CAL intervention started on October 23rd and concluded on May 3rd.[14] During the first week of implementation, all treated students took a mandatory "Initial Knowledge Check" through the learning platform (this is the first activity in the platform). Based on the information from this initial assessment, the platform predicts the list of topics students master, do no yet master, and are ready to learn, and creates an individualized learning path for each student.

The implementation of the CAL component faced several challenges that affected the delivery of some sessions. The most prominent issues included unstable internet connectivity, insufficient availability of computers, frequent power outages, and unplanned school activities that interfered with the scheduled sessions.[15] To mitigate the loss of sessions caused by connectivity problems, wireless modems were provided to supervisors. Additionally, some schools struggled to provide enough devices for all students during the CAL sessions. To address this shortfall, tablets were allocated to supervisors for use in these schools. Another challenge was the frequent occurrence of school events and social activities, which were often unplanned or not accounted for in the school calendar. These events sometimes overlapped with the scheduled CAL sessions, leading to the cancellation or rescheduling of sessions and further disrupting the learning schedule.

3.3 Implementation: Tutoring Component

The first tutoring session took place on October 23rd and the last on April 26th. Tutoring sessions took place one to three days per week during extracurricular hours. In all but one school, tutoring sessions took place during hours allocated for optional workshops.[16] Optional workshops were meant to cover extracurricular content and varied in scope. Some examples of the type of courses covered during these hours were computing, music or dance lessons. The workshops and tutoring sessions would typically take place between 2 and 3:30pm. In total, 61 students were selected to participate in these tutoring sessions.

The implementation of the tutoring component also faced several challenges. School activities frequently disrupted the tutoring schedule, and unlike the CAL sessions, these missed tutoring sessions could not be rescheduled.[17] Additionally, classroom availability posed a significant issue, resulting in tutoring sessions being held in the school cafeteria or other open public spaces where

[13] Students in control classrooms received temporary licenses to access the software approximately 2 months after the endline data collection.

[14] This is equivalent to roughly 22 weeks, excluding winter, Holy Week, and two weeks where the use of the platform was interrupted due to training and data collection.

[15] These issues are common across schools in the DR due to inadequate infrastructure.

[16] One of the nine schools did not offer optional workshops, so students stayed after school hours to receive tutoring sessions.

[17] If a class missed a CAL session, teachers often reallocated other math hours that week to make up for the missed CAL session.

noise and distractions were prevalent. Another challenge tutors encountered was sustaining student interest and participation. Some students were reluctant to join the tutoring sessions, leading to delays in starting the activities.[18]

4 Data

To evaluate the effectiveness of the program, we assess several key outcomes: students' academic performance, socioemotional skills, psychological well-being, and time use both inside and outside of school. Additionally, to better understand the potential drivers behind any observed effects, we examine changes in teachers' resource allocation based on student-reported information. These outcomes were collected using a combination of survey data, administrative records, data from the adaptive software, and brief math assessments. Each of these outcomes is described in detail below.

Student surveys and math assessments were administered at baseline, midline, and endline by the research team. Baseline surveys were collected in September 2023, followed by midline surveys in December and endline surveys in May 2024. Administrative records were collected and digitized after the end of the school year, between June and September 2024.

Academic performance. Our main outcome of interest is students' academic performance, as we want to examine whether the platform or tutoring component were effective in improving student learning in math. To do this, the research team administered math assessments at baseline, midline, and endline. These assessments were designed using items of varying difficulty levels, adapted from MINERD's national assessments. Endline tests were designed by a pedagogical expert and followed the content and cognitive areas used on the 2019 Trends in International Mathematics and Science Study (TIMMS) 8th grade math assessments, which included seven content domains and three cognitive areas (Mullis and Martin, 2017).[19] We examine student performance in each of these content areas and cognitive domains, both at midline and endline. Moreover, we classify items according to their difficulty levels to assess differences in impacts by the grade level of the content.

In addition to the tests, between June and September 2024 we collected and digitized schools' administrative records containing data on students' end-of-year school grades for math and Spanish (to test for adverse effects), their attendance records, whether they dropped out during the school year, and whether they passed or failed the grade.

Socioemotional skills. As students increasingly rely on digital devices for learning and have fewer face-to-face interactions with their teachers, there is a risk that socioemotional or non-cognitive skills may be negatively impacted. In baseline and endline surveys we measured the following three skills: perseverance, locus of control, and grit. Perseverance is measured through a real effort task adapted from Alan et al. (2019). In this task, students first answered a logic question, and then chose one of three options for their next step: attempting another question of the same difficulty, tackling a more difficult question, or giving up. Their choice is used as our measure of perseverance. To measure locus of control, or the extent to which students believe they have control over the outcome of events in their lives, we used an adaptation of Rotter's locus of control scale (Rotter, 1966). Lastly, we measured grit using a 4-item scale adapted from Carlana

[18] If a large number of students were absent from the sessions but present at school, the tutor often spent several minutes at the beginning of the session trying to locate them.

[19] The 8th grade TIMMS math assessments evaluated the following content domains: (1) Numbers, (2) Algebra, (3) Geometry, and (4) Data and Probability. The cognitive domains evaluated were: (1) Knowing, (2) Applying, and (3) Reasoning. TIMMS is only administered in the 4th and 8th grades.

and La Ferrara (2021). We build indices for the grit and locus of control measures such that the minimum and maximum values are 0 and 1.

Student well-being. The nature of the learning platform, in which each student works independently, may have led to less opportunities for teamwork or support than regular instruction, potentially making students feel more isolated in class. Moreover, increased screen time has been linked to lower psychological well-being among teenagers (Stiglic and Viner, 2019; Twenge and Campbell, 2018). We used questions from the Children's Depression Screener (ChilD-S) developed by Frühe et al. (2012) to measure students' well-being.

Furthermore, by increasing students' math skills, or their awareness of their math skills, access to CAL may have mixed impacts on math anxiety. Following Araya et al. (2019), we created a math anxiety index using students' answers (on a Likert scale) to the following statements: 1) I get nervous before math tests and 2) I am worried of having bad grades in math.

Teacher behavior. The CAL platform may free up teachers' time by automating some of their daily tasks, such as developing and grading homework and tests. As a consequence, we may observe differences in the allocation of teachers' time and effort across classroom activities. To assess changes in these outcomes, we ask students about their perceptions of teacher behavior in the classroom (e.g. did the teacher give you additional support outside of the classroom?, did they clarify a lesson that you did not understand?) and their time management (e.g. did the teacher start class late?, did the teacher come to school?). We use their answers to build four indices of teacher behavior: 1) time mismanagement index, 2) teacher engagement index, 3) teacher effectiveness index, and 4) teacher ineffectiveness index. The first index, on time mismanagement, is based on students' answers to a series of questions on teachers' timeliness to class and school. The third index is built upon students' answers to a set of questions around their engagement during class.[20] The last two are based on classroom activities, which we classify into "effective" and "ineffective" practices, depending on their correlation with control group students' midline and endline test scores. Practices that are positively correlated with test scores are combined into an "effective practices" index and those negatively correlated are used for an "ineffective practices" index.[21] All indices are standardized to have a mean zero and a standard deviation of 1 in the control group. The questions used have been adapted from questionnaires used by De Hoyos et al. (2021) and Ferguson and Danielson (2015).

Software data is used for both monitoring and assessment purposes. The software contains information on students' interactions with the software, including the number of sessions held, the date and duration of each session, and the number of newly mastered topics by each student. We use this information to assess compliance with the intervention and to measure progress within the treatment group. In addition to this, we use the software data to measure the frequency with which teachers use it to assign and grade homework, in-classroom activities, and assessments. This data is available for the treatment group only.

[20]The questions used to build this index are: 1) Your math teacher keeps students engaged, 2) Asks questions to check for understanding, 3) Knows when everyone understands, 4) Makes every student work hard, 5) Assigns homework that helps you learn, 6) Reviews past topics, 7) Marks your work to help understanding, and 8) Wants you to explain your answers.

[21]The "effective" practices are: 1) You used a textbook, 2) You copied from the board or book, 3) You solved problems, and 4) You practiced problems on the board; while those classified as "ineffective" are: 1) You worked in a group, 2) Your teacher explained math topics, and 3) Your teacher asked you to practice for a test.

4.1 Balance Checks and Attrition

We present balance checks in Table 1 using information from baseline surveys. There are no statistically significant differences between the treatment and control groups across observable characteristics. Approximately 56 percent of the sample is female. The average age of students in our sample is close to 15 (the typical age for a students enrolled in 9th grade is 13 to 14 years old), reflecting the high levels of grade repetition in the country (approximately 19 percent of students in our sample repeated at least one grade). Baseline scores are relatively low: students answered an average of 6 out of 12 questions correctly. Moreover, we can observe that the average student in our sample comes from a low socioeconomic background; for instance, less than 40 percent of the sample has a mother or father who attended college.

As shown in Columns 1 and 5 of Table 2, survey and test completion rates were high at both midline and endline: 92 percent of the baseline sample completed surveys and tests at midline, and 90 percent did so at endline. The differences in completion rates between the control and treatment groups are small: 0.5 percentage points at midline and 2.9 percentage points at endline. While the latter difference is statistically significant at the 5 percent level, it is small enough not to raise substantial concerns.

Most importantly, the students who completed the surveys appear similar to those who did not. Columns 2-4 and 6-8 show minor differences in observable characteristics between respondents and non-respondents. For instance, treatment group students who responded to endline surveys were, on average, 0.037 years younger (less than half a month) than non-respondents, and the proportion of female respondents was 4 percentage points lower. Baseline test scores were 0.01 standard deviations higher among respondents in the treatment group. Overall, we believe the low rates of attrition and minimal differences between respondents and non-respondents do not compromise the integrity of our findings. In Appendix Table A4, we present Lee (2009) bounds for the main performance outcomes. The results indicate that selection due to differential attrition does not account for the observed effects, as these appear similarly large after applying this bounding method, with gains of 0.09-0.32 standard deviations at midline (not statistically significant) and 0.23-0.37 at endline for the CAL-only treatment. The tutoring interaction remains statistically insignificant.[22]

5 Empirical Framework

We estimate the impact of the program on student outcomes using the following regression model:

$$Y_{ci} = \beta_0 + \beta_1 \text{CAL}_c + \beta_2 \text{Tutoring}_c + \beta_3 X_{ci} + \epsilon_{ci}$$

where Y_{ci} is the outcome of student i in classroom c and CAL_c is a binary variable that takes the value of 1 if classroom c was assigned to treatment and 0 otherwise. $Tutoring_c$ takes the value of 1 if the classroom was assigned to the group tutoring treatment and 0 otherwise. X_{ci} is a vector of baseline characteristics for student i in classroom c. In this specification, β_1 captures the average treatment effect (ATE) of the CAL software, while β_2 captures the additional impact ("value-added") of the tutoring component relative to classrooms that received only the CAL software. In the results section we present each coefficient separately and the sum of the two. The latter

[22]We calculate the lower bound by dropping the fraction of highest scoring test-takers in each treatment group until the response rates are equal across the three experimental groups. To estimate the upper bound, we drop instead the fraction of lowest-scoring students. We do the opposite when response rates are highest in the control group.

represents the effect for classrooms assigned to both the CAL and tutoring interventions, relative to the control group.

Throughout this paper, we use randomization-based inference as our preferred methodology for analysis, which has gained prominence over traditional sampling-based methods in recent years (Athey and Imbens, 2017; Green et al., 2011; Young, 2019). This approach leverages the experimental design to derive statistical inferences directly from the randomization procedure, ensuring that our methods align closely with the study's design. Randomization inference is particularly advantageous in the context of nonstandard randomization techniques, such as the pairwise randomization used here, as it allows us to account for the specific structure of the randomization. Moreover, it enables us to obtain nearly exact p-values without relying on assumptions about the shape of the sampling distribution, which is especially helpful in small samples or when the number of clusters is limited, as is the case in this study. In such situations, traditional assumptions about the sampling distribution may be less reliable, making randomization inference a more robust approach.

To test the sharp null hypothesis of no treatment effect, we use a potential outcomes framework. In this framework, the potential outcome for each individual i in classroom c depends on two factors: the classroom's assignment to the CAL software (Z_c) and the classroom's assignment to the tutoring intervention (W_c):

- Z_c is a binary variable indicating whether classroom c was assigned to receive the CAL software ($Z_c = 1$) or not ($Z_c = 0$).
- W_c is a binary variable indicating whether classroom c was assigned to the tutoring intervention ($W_c = 1$) or not ($W_c = 0$). Note that within classrooms assigned to the tutoring intervention, only 20 percent of students received the actual tutoring, but the assignment W_c was made at the classroom level.

The potential outcomes are defined as follows:

- $Y_{ci}(1,1)$ represents the outcome for individual i in classroom c if the classroom is assigned to receive both the CAL software ($Z_c = 1$) and the tutoring intervention ($W_c = 1$).
- $Y_{ci}(1,0)$ represents the outcome for individual i in classroom c if the classroom is assigned to receive the CAL software ($Z_c = 1$) but not the tutoring intervention ($W_c = 0$).
- $Y_{ci}(0)$ represents the outcome for individual i in classroom c if the classroom is assigned to the control group ($Z_c = 0$), with neither the CAL software nor tutoring.

The sharp null hypothesis posits that these potential outcomes are identical regardless of the treatment condition, meaning the treatment has no effect. Formally, this can be stated as:

$$Y_{ci}(1,1) = Y_{ci}(1,0) = Y_{ci}(0) \quad \text{for all individuals } i \text{ in all classrooms } c$$

This implies that any observed differences in outcomes between the groups are due solely to the random assignment process and not to the intervention itself. By simulating 2000 randomizations, we approximate the sampling distribution of the ATE under the sharp null and estimate the probability of observing an ATE at least as large as the one found in the actual experiment if the sharp null were true. This simulation process is repeated for each outcome analyzed.

The validity of our findings relies on the "no-interference" or Stable Unit Treatment Value Assumption (Rubin, 1978), which assumes that the treatment of one classroom does not affect the potential outcomes of students in another. The platform used in this study required each student to log in using a unique username and password, and its use was closely monitored during the designated period for this task. These measures minimize the likelihood that students in non-treated classrooms could have accessed or benefited from the platform, thereby supporting the assumption of no interference. However, it is conceivable that treated students may have interacted with control students in ways that influenced their outcomes (e.g., through assistance with homework or exam preparation). Additionally, the assignment of a classroom to treatment may have impacted how teachers interacted with students in non-treated classrooms. We do not have anecdotal evidence suggesting that teachers changed how they interacted with control group classrooms as a result of the treatment, or that treated students collaborated with peers from different classrooms (based on conversations with the teaching team).

6 Results

6.1 Tutoring and CAL Use

The results from the initial CAL assessment suggest that, at the time of completion, the median student mastered only 2.7 percent (9) of the 337 math topics included as part of the course. Throughout the school year, the median treated student spent approximately 14 hours (approx. 38 minutes per week) on the platform and learned an average of 45 new math topics through it. As shown in Figures 4 and 5, the distribution of the number of hours and the number of topics learned on the platform is left-skewed, with the top 10th percentile of students spending over 38 hours using the platform and learning more than 188 new math topics throughout the school year.

To examine whether students used the CAL platform outside of school, we assess the share of time where the student used the software outside of the scheduled days. Note that this is a conservative estimate, given that teachers had flexibility to reschedule the use of CAL if classes had been cancelled during the scheduled days. Using this proxy, we find that for the median student 15 percent of the time spent using the platform took place outside the regularly scheduled days. This is equivalent to approximately two hours total. Given these numbers, we safely conclude that any effects observed come from a substitution of the time spent during class and not from additional instruction time.

Of the 61 students assigned to the tutoring treatment arm, all received at least 2 hours of tutoring, with the median student receiving 29 hours, or approximately 1.4 academic hours per week. Note that this is significantly below the intended dosage of 3 hours per week. This is partially explained by student absenteeism during the tutoring sessions. On a given week, up to 47 percent of students failed to show up to at least one of the meetings, based on monitoring data. We include a more detailed discussion of this and other implementation challenges in section 1.7.

6.2 Program Effects

6.2.1 Student Performance

We present Intention-to-Treat (ITT) effects roughly two (midline) and six months (endline) after the start of the program in Table 3. Columns 2 and 3 present results at midline and columns 4 and

5 at endline. We find that by midline the CAL-only treatment had produced gains of approximately 0.17-0.21 standard deviations, as shown in Columns 2 and 3. Although large, these results are not statistically significant. These impacts are roughly consistent across specifications (with or without covariates). We do not find significant effects from the tutoring treatment at midline, likely because students had only received 8 tutoring sessions at this stage.

By endline, CAL-only effects had increased to 0.29-0.31 standard deviations (statistically significant at the 5 percent level), as shown in Columns 4 and 5. The coefficient associated with tutoring was negative and relatively large at -0.19 when including covariates. Although not statistically significant, the magnitude of the coefficient is a cause for concern and we explore in the following section some hypotheses behind these results. We believe this negative coefficient may partly reflect disruptions to the tutoring sessions due to scheduling conflicts with other school activities and a lack of student engagement (including high levels of absenteeism). Additionally, tutor availability issues resulting from weather and transportation challenges likely further impacted the consistency and quality of tutoring sessions. Effects on performance from the CAL component align with those found in other work using computer-assisted instruction in low or middle income countries.[23]

When looking at school grades by the end of the year, we find small negative differences on students' math grades for CAL-treated students, relative to the control group; however, these differences are not statistically significant (see Table 8). We present effects on Spanish as a robustness check for the previous result. Interestingly, we do not observe similar differences for Spanish grades. The observed patterns hold whether we condition grades on having finished the school year (Panel B) or use the average of available periods for students who dropped out and/or are missing information for at least one school period (Panel A).[24]

6.2.2 Intrinsic Motivation, Absenteeism, and Dropout

Beyond academic performance, improvements in learning could also positively influence students' retention in school by increasing their motivation and interest in the subject, as well as their chances of obtaining a passing grade. In Tables 4-7, we present impacts on intrinsic motivation, preferences towards math, absenteeism, and student dropout. Student dropout is a significant concern in our context, especially among the targeted age group. In the Dominican Republic, over 40 percent of students drop out between the 8th and 12th grades (IDEICE, 2017). In the control group, 8 percent of students dropped out of school during the intervention period (see Table 7). However, as shown in Tables 4-7, we find no evidence that the intervention impacted students' intrinsic motivation, how much they like math (self-reported 1 to 4 scale), school absenteeism, or dropout rates by endline. The lack of findings on intrinsic motivation align with Araya et al. (2019), who also observed no effect on this outcome in a study using a game-based learning platform in Chile.

6.2.3 Math Beliefs and Self-Concept

As students spend more time on the software and master new topics, they receive new information on their abilities and may update their beliefs accordingly. In Tables 9 and 10, we present effects

[23] For example, Araya et al. (2019) report effects of 0.21-0.27 standard deviations (SD) from a computer-assisted, game-based math learning platform in Chile after seven months of program exposure. In India, Muralidharan et al. (2019) find effects of 0.37 SD after 4.5 months. Finally, Büchel et al. (2022) observe effects of up to 0.17 SD from a computer-assisted learning software in El Salvador over a six-month period.

[24] The school year is divided in four quarters. Final grades are estimated as the average of each quarterly grade.

on students' beliefs about their math performance (expected test scores), their confidence in the subject, and their math self-concept. Panel A of Table 9 shows that CAL students expected higher test scores than the control group (by 0.12-0.13 SD). However, this gap is not statistically significant and is not as large as the one observed in actual test scores (0.3 SD), suggesting that this group may have downwardly biased beliefs about their endline performance.

We do not find differences in expected performance between CAL-only classrooms and those that received tutoring, but Appendix Table A1 shows that students in the latter group were more likely to be overconfident in their performance, although these results are not statistically significant.[25] Despite the lack of statistical significance, the differences in magnitudes suggest a potential for overconfidence in the CAL+Tutoring group and underconfidence in the CAL-only treatment. Panel C in Figure 3 illustrates the gap between students' expected performance and their actual performance on endline math tests, expressed as a share of correct answers. The largest gap is observed in the CAL+Tutoring group, followed by tutees and the control group. The smallest gap observed corresponds to the CAL-only group. These findings mirror those of Table 9 (Panel B), which shows that the CAL-only and CAL+Tutoring groups had similar scores on a math self-confidence index, built upon a series of questions on how comfortable they feel learning and applying math.[26]

Lastly, we examine math self-concept, defined as students' perceptions of their own ability in mathematics compared to others (see Table 10). As in Araya et al. (2019), despite the large increases observed in math performance, we do not find statistically significant differences across experimental groups in a math self-concept index or on individual measures used to build the index. However, students in the CAL+Tutoring group were 9.3 percentage points less likely to agree with the statement "Math is harder for me than for many of my classmates" than students in the control group or in the CAL-only group. While these results are not conclusive, they may be suggestive of shifts in students' self-perception and confidence in their math abilities. Despite performing worse on math tests, the tutoring group seems to feel equally confident about their performance and are more likely to have mismatched expectations about their outcomes in the test, when compared to those whose classrooms were not assigned to the tutoring component.

6.2.4 Self-Efficacy, Grit, Perseverance, and Locus of Control

By offering personalized learning pathways and continuous feedback on individual progress and milestones, the platform allows students to understand and learn from their mistakes privately, without the stigma of public correction. As students master new topics and receive real-time feedback, this process may enhance their sense of competence and self-efficacy. In this subsection, we examine four related outcomes: self-efficacy, grit, perseverance, and locus of control. As shown in Table 12, neither the CAL intervention nor the tutoring intervention had a significant impact on a self-efficacy index.[27] This suggests that the interventions did not notably alter students' beliefs

[25]The overconfidence outcome takes the value of 1 if students' estimated performance on the math test exceeds their actual performance.

[26]The outcome variable for the self-confidence table is a standardized index based on students' answers to the following items: 1) When teachers teach me a new math topic, I learn it without much difficulty; 2) When I take a math test, I am confident that I can answer the questions well; 3) I can help my classmates with math homework that they don't understand; and 4) I feel more capable as I learn more things in math.

[27]This index is built using students' answers to the following items: 1) I can always solve difficult problems if I try hard enough, 2) I can solve most problems if I invest the necessary effort, 3) I can stay calm when I face difficulties because I can trust my coping skills, 4) When faced with a problem, I can usually find several solutions, and 5) If

in their ability to achieve academic success.

Similarly, we do not find impacts on a measure of grit (see Table 13), which assesses students' perseverance and passion for long-term goals (Duckworth et al., 2007). Following Alan et al. (2019), we implement an alternative measure of perseverance, based on a real-effort exercise. Table 14 shows that treated students are 8.7 percentage points more likely to choose to undertake an easier task against a more challenging alternative. We do not observe differences in their likelihood of giving up after failure for either group. This poses the possibility that the adaptive nature of the platform, which presents exercises based on students' current ability, may have negatively impacted their willingness to take on more challenging exercises.

Lastly, we examine effects on locus of control, which refers to the degree to which individuals believe they have control over the outcome of events in their lives. A higher internal locus of control indicates that individuals feel they can influence their own success, while an external locus of control suggests that they attribute outcomes to external forces. Table 15 shows that the CAL intervention had no significant impact on students' locus of control. However, the tutoring component had a small but statistically significant negative effect on this outcome (0.02–0.03 percentage points), indicating that students in the tutoring group were slightly more likely to feel that their performance was determined by external factors rather than their own actions. We include a discussion on this point in the following section.

6.2.5 Student Wellbeing

To assess changes on students' wellbeing, we use two proxies: the Children's Depression Screener (ChilD-S) and a a short questionnaire on math anxiety. For the first one, we build an index based on students' answers to 8 items about their mental state (e.g., "I feel happy", "I feel lonely", "It is hard for me to concentrate"). For the second one, we use two questions on students' worry about their math grades and their feelings before math tests to build a math anxiety index. Table **??** shows that the intervention had no impact on the first measure of wellbeing. While we do not find impacts on a math anxiety index, we find important differences when looking at individual measures of math anxiety. As seen in Table 18, students in the CAL-only group were 6 percentage points more likely to get nervous before math tests than students in the control and CAL+Tutoring groups. This difference is statistically significant at the 5 percent level. While math anxiety is typically linked to lower performance (Dowker et al., 2016), this result may be positive if it signals higher levels of awareness or concern about their math performance in this group.

In the Appendix, we present results on gender beliefs (Table A3) and student time use (Table 16). We do not find statistically significant effects on these outcomes.

6.2.6 Teacher Practices and Time Use

In this section, we estimate the effects of the intervention on teachers' allocation of time and effort across various activities, as this is one potential mechanism by which the intervention may have impacted teacher effectiveness and student performance. In Panel A of Table 19, we show impacts on a time mismanagement index based on students' responses to a series of questions on teachers' timeliness to school/class.[28] We do not find statistically significant differences on this index across

I'm in trouble I can find a solution.

[28]The questions used to construct this index are: 1) Was your teacher late for class?, 2) Did they not come to school?, 3) Did they start the class early?, 4) Did they finished class early?, 5) Did they finished the class late?, and

treatment groups. In Panel B we assess effects on an index of teacher engagement, built upon students' answers to questions around their engagement during class.[29] Although the index is lower for treated students, relative to the control group, these differences are not statistically significant.

Lastly, we look at classroom activities, which we classify into "effective" and "ineffective" practices, based on their correlation with student midline and endline test scores. Practices that are positively correlated with test scores are combined into an "effective practices" index and those negatively correlated are used for an "ineffective practices" index. As shown in Table 20, we do not find statistically significant differences across experimental groups on these outcomes.

Overall, the lack of findings on teacher behavior suggest that the effects observed may be attributed to the substitution of some of the teachers' instructional time with the new technology (skill replacement), rather than a reallocation of teachers' time and effort (or that of their students) across different tasks. This is also consistent with our analysis of platform data. Although teachers spent an average of 33 minutes per week actively using the platform, they seldom leveraged it to automate routine tasks such as assigning or grading homework and exams. Our data indicates that, over the course of the academic year, teachers assigned no more than three activities through the CAL platform.

6.3 Heterogeneity in Program Effects

As students progressed through the software, they were targeted with exercises that catered to their learning levels. Following Muralidharan et al. (2019), we classify test topics into two categories–below and at grade level–and test whether students fare differently across these two categories in the tests. Table 21 suggests that the CAL treatment was effective in improving performance on both below and at grade level content, with impacts ranging from 0.21 to 0.28 standard deviations. This pattern differs from that found in Muralidharan et al. (2019), which finds large impacts on below grade math items but no impacts on at grade level items.[30] However, the patterns are consistent with what we would expect by looking at software use, as the software was designed to cover both below and at grade level content. At midline, when students had spent most time on modules covering below grade level topics, we find that effects were largely concentrated on test items covering this type of content.

In Table 22, we present effects across content areas included in TIMMS mathematics assessments for 8th grade. We observe positive differences from the CAL treatment on the algebra, numbers, and geometry content areas by endline, although the coefficient for geometry is not statistically significant. We do not find impacts on data and probability, the one TIMMS content area that was not covered in the platform or in the Dominican Republic's 9th grade curriculum. For the tutoring component, results seem to mirror those found on Table 3, as classrooms assigned to a tutor show negative but statistically insignificant differences, when compared to the CAL-only treatment, across all content areas except for data and probability. Beyond content areas, TIMSS also evaluates students' abilities in three cognitive domains: knowing, applying, and reasoning. In Table 23, we present results across these cognitive areas. As shown, we find large and positive

6) Did they left school early?

[29]The questions used to build this index are: 1) your math teacher keeps students engaged, 2) Asks questions to check for understanding, 3) Knows when everyone understands, 4) Makes every student work hard, 5) Assigns homework that helps you learn, 6) Reviews past topics, 7) Marks your work to help understanding, 8) Wants you to explain your answers.

[30]In Hindi, the authors find significant effects for questions at and below grade level.

effects on all cognitive areas, ranging from 0.16 to 0.31 standard deviations, although results are not statistically significant for the "knowing" cognitive area.

Lastly, Figure 6 presents the distribution of performance for both the treatment and control groups across baseline performance percentiles. Although confidence intervals overlap at several points, we observe that the treatment group consistently outperformed the control group at all levels of baseline performance.

7 Discussion

In this section, we discuss how the implementation challenges we faced when deploying CAL in schools could have affected the results observed on performance, and delve into a key remaining question: why was the tutoring component ineffective? While we do not have conclusive evidence to explain these findings, we present in this section some potential explanations.

7.1 CAL Implementation Challenges and Impacts on Take-up

Challenges mentioned in subsection 3.2 resulted in lower-than-expected use of the CAL platform, with the average student using the platform for less than 60 percent of the intended time. In Appendix Table A5, we estimate the dose-response relationship between hours of CAL use and value-added (Muralidharan et al., 2019). We instrument for hours of use using the random assignment to the experimental treatment arms. The IV estimates indicate that, on average, one additional hour of CAL use increased endline test scores by 0.027 standard deviations. Figure A2 shows that diminishing returns start to kick in around the 1500-minute mark. Up until that point, the slope of the curve increases steeply, but then the rate of increase in math test scores begins to taper off, signaling that additional time on CAL has progressively smaller benefits on test scores. Note that the effect identified here averages causal effects over varying levels of treatment. Thus, an extrapolation of up to 1500 minutes of use is likely to be downwardly biased, whereas an extrapolation of more than 1500 minutes of use may overstate the true effects of the intervention. In Table A5 we limit our extrapolation to 1500 minutes of use, which, assuming 80 percent attendance, would be equivalent to roughly 5 months of implementation.[31] Our results suggest that implementing the program with fidelity for 5 months would lead to test score gains of 0.73 standard deviations.[32]

7.2 Ineffectiveness of Tutoring Component

The challenges mentioned in subsection 3.3 help explain the limited effectiveness of the tutoring sessions. By midline, the average attendance at tutoring sessions was approximately 75 percent. In an effort to address this, students were given the option early in the spring to withdraw from tutoring and join an alternative extracurricular workshop if they preferred. While the majority (54

[31]The large majority of treated students in our sample (approx. 90 percent) fall below the 1500-minutes mark, which reduces our concern that diminishing returns may bias these estimates. Moreover, we cannot reject a test of equality (difference-in-Sargan) between the IV and OLS estimates using a value added specification, and as seen across columns 1-3, the coefficient associated with hours of use is similar across model specifications, which suggests that the LATE may be similar to the ATE.

[32]This extrapolation requires the assumption of heterogeneity in treatment effects. As shown in Figure 6, we do not find evidence of effect heterogeneity across baseline performance.

out of 61 students) remained enrolled in the tutoring program through the end of the school year, 7 students opted to withdraw at the beginning of the spring semester. Despite the relatively low dropout rate, attendance continued to pose a challenge, with average student attendance standing at 73 percent by the end of the intervention. Using a meta-analytic sample of 265 tutoring studies, Kraft et al. (2024) find that program features that change as tutoring programs are taken to scale, including higher tutor-tutee ratios and shorter delivery dosages, partially account for the reduced effectiveness of these programs (when compared to smaller scale interventions).[33]

Moreover, if the program disrupted other productive learning activities without delivering the consistent support needed to drive academic improvement, it could have had a net negative impact. While no observable differences were found between the tutoring group and the control group, we did observe negative (though not statistically significant) differences when compared to the CAL-only group, as well as a reduced usage of the CAL platform. This issue is explored in greater detail in the following subsection.

1. Use of CAL platform. We find that students in classrooms assigned to the tutoring arm spent less time on the platform and put lower effort while using it.[34] Figure 9 shows that the average student assigned to a CAL-only classroom spent a total of 1232 minutes using the platform (56 minutes per week) while the average student in a CAL+Tutoring classroom spent 1082 minutes (49 minutes per week). We use the relationship between minutes of use and test scores to predict how this difference in use may affect results. Assuming a linear dose-response relationship between the two, we find that this difference in time use may account for approximately 34 percent (0.07 SD) of the observed difference in endline performance between classes with tutoring and those that only included the CAL component (see Appendix Table A5). Within CAL+Tutoring classrooms, both non-tutees and tutees had considerably lower levels of use than the average student in a CAL-only classroom, but the difference in use was significantly lower for tutees, who spent only 1000 minutes (or 45 minutes per week) using the platform.

2. Selection of Tutees. Although we anticipated that teachers would select the lowest-performing students as tutees, the data show otherwise. The average tutee was not in the bottom 20th percentile on baseline tests; in fact, as Figure 7 illustrates, the average tutee was located in the second quintile of performance, approximately 0.7 standard deviations below the average. This discrepancy may stem from the timing of the selection process, which occurred at the start of the school year when teachers may have had limited knowledge of their students' abilities. A plausible explanation is that initial tutee selection was based on incomplete or inaccurate information, or that teachers considered students' likelihood of attendance as one selection criteria, perhaps assuming that the worst performing students were less likely to attend the tutoring sessions.[35] It remains unclear whether teacher judgment or baseline test scores should serve as the optimal criteria for selecting tutees. Nevertheless, as Figure 8 shows, students selected as tutees were closer to the mean than to the bottom, and exhibited flatter growth in test scores between the baseline and endline than other comparable groups, suggesting that the selection process may not have effectively targeted

[33] Kraft et al. (2024) show a declining pattern of effects as student-tutor ratios increase.

[34] We define effort as the ratio of topics learned to time spent on the platform.

[35] After the initial selection, we provided teachers with the list of the bottom 20 percent of students in their class, along with their baseline test scores. While 70 percent of teachers made at least one adjustment to their tutee list, only 60 percent of the students from the original list were changed. Only 41 percent of the students on the updated lists were from the bottom 20 percent in terms of baseline performance (an increase of 20 percentile points compared to teacher's initial selection). The most frequently cited reason for selecting a tutee was having low performance (82 percent), followed by showing behavioral issues (6 percent) or lacking an appropriate study method (6 percent).

the students who would have benefited the most from the tutoring intervention.

3. Locus of Control. Students in the CAL+Tutoring group were less likely to score positively on the locus of control variable (Table 15). The significance of locus of control becomes apparent when we examine the effects by students' baseline levels on this variable. When the sample is split into high (above the baseline mean) and low (below the baseline mean) locus of control groups, Table 24 shows that the large positive effects observed in the CAL-only group are concentrated among students with initially high levels of locus of control, while the negative effects in the tutoring group are concentrated among those with lower initial levels on this variable. A possible explanation is that the presence of a tutor may have led students to perceive them as responsible for their academic success, making them more confident about their outcomes and more passive in their efforts. This could explain the lower use of the CAL platform among students in tutoring classrooms, as they might have relied more heavily on the tutor's support. In Table 18, we show that students in the CAL-only group were more likely to report getting nervous before math tests than students in the control and CAL+Tutoring groups.

8 Conclusion

This paper contributes to the growing literature on computer-assisted learning by testing an innovative intervention that replaces nearly one-third of regular math instruction with the use of an adaptive learning platform. Conducted in the Dominican Republic, a lower-middle-income country, this study highlights both the challenges and transformative potential of introducing technology-based tools in public school classrooms. We find that substituting two weekly hours of regular instruction with CAL use significantly improves student performance on math tests by up to 0.3 standard deviations, with these gains being consistent across various difficulty levels and subject areas.

Interestingly, when the CAL component was combined with a tutoring intervention targeted at 20 percent of students in each classroom, the effectiveness of the CAL component was reduced, even for those who did not directly participate in the tutoring sessions during extracurricular hours. Implementation challenges, such as low attendance rates among tutees and reduced platform usage and effort among students in the tutoring group, may explain this diminished effectiveness. Although we do not have conclusive evidence for why platform use decreased, we find impacts on student locus of control and wellbeing, with students in the tutoring group exhibiting lower locus of control and lower math anxiety compared to those who only received the CAL component. While these results do not fully account for the reduced performance observed in the tutoring group, they underscore the need to better understand how technology-based interventions interact with other classroom supports.

Overall, this paper illustrates the promising potential of technology to enhance education systems worldwide. However, it also highlights the importance of understanding the dynamics between technology integration and concurrent interventions, such as tutoring, within the school context. Furthermore, we highlight the importance of infrastructure and logistical planning in the successful implementation of programs that take place in schools and rely on the use of technology, underscoring the need for continuous monitoring and adaptive strategies to minimize disruptions. As part of future work for this project, we plan to conduct qualitative research, including focus groups and interviews with teachers and students, to explore potential shifts in attitudes or behaviors resulting from the tutoring treatment arm.

Figure 1: Timeline for Intervention and Study

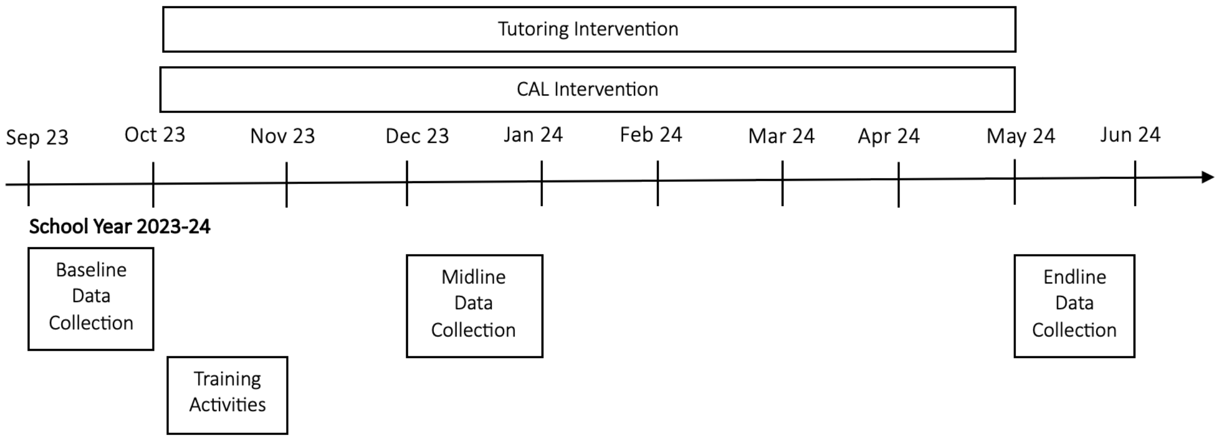
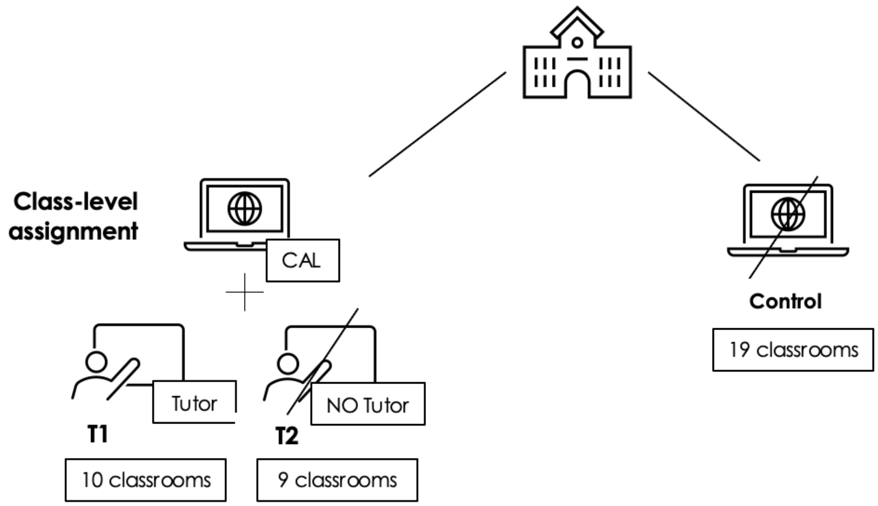
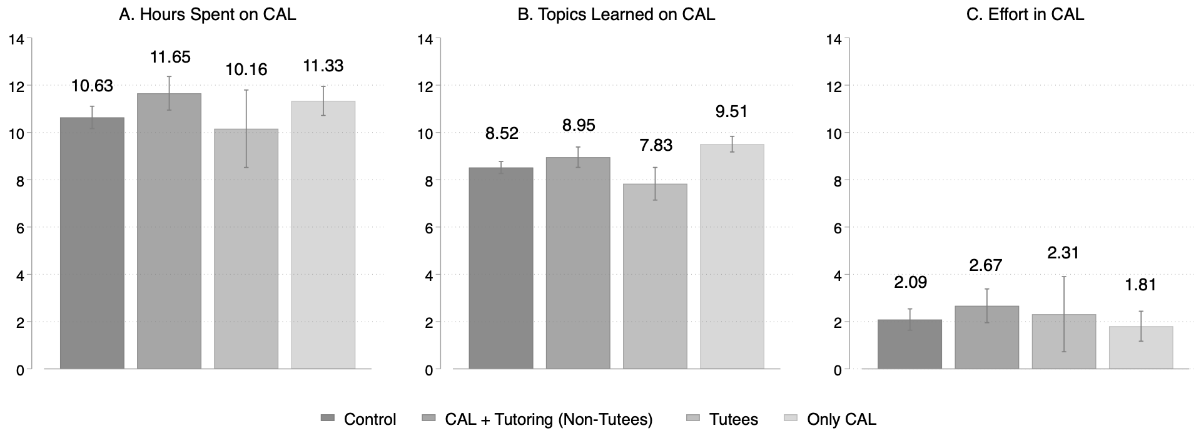


Figure 2: Randomization Design



Notes: T2 is composed of 10 classrooms assigned to receive access to the CAL software. T1 is composed of 9 classrooms where in addition to receiving access to the software, the bottom quarter of students, in terms of performance, were to receive tutoring sessions in small groups during extracurricular hours. The control group, containing 19 classrooms, did not receive access to the CAL or tutoring components.

Figure 3: Expected Performance on Endline Math Tests



Notes: The CAL+Tutoring group excludes tutees. Panel C shows the difference between students' reported expected performance (share of correct answers) and their actual performance.

Figure 4: Time Spent on CAL

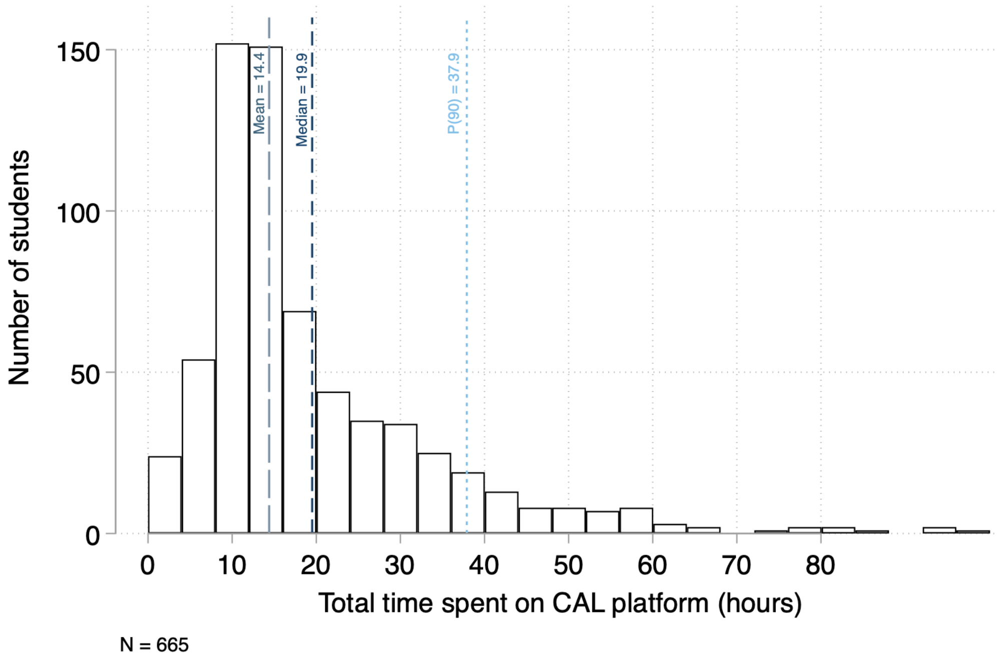


Figure 5: Topics Learned on CAL

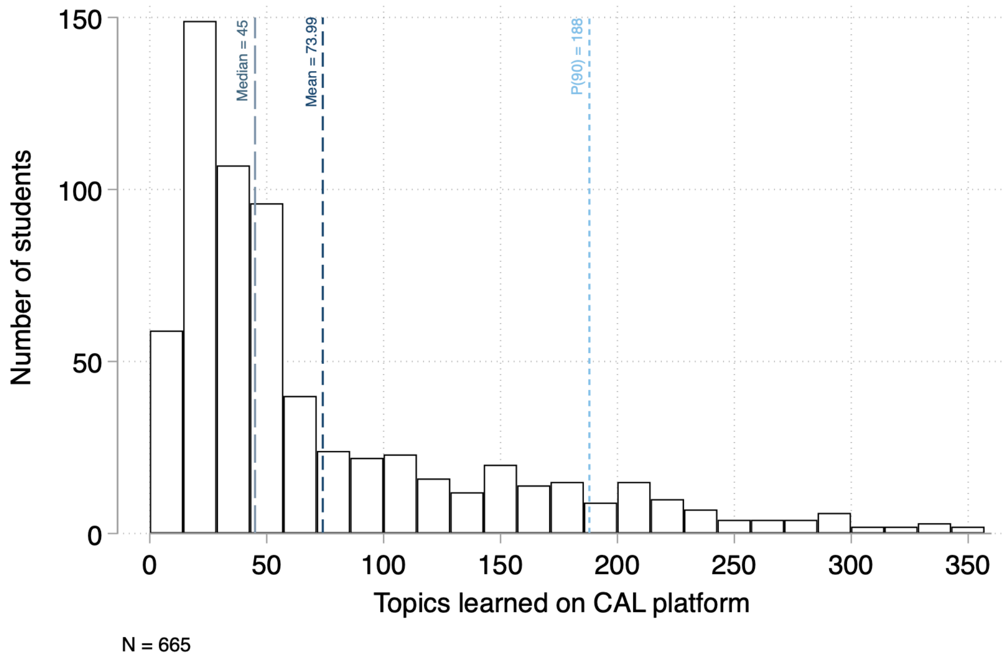
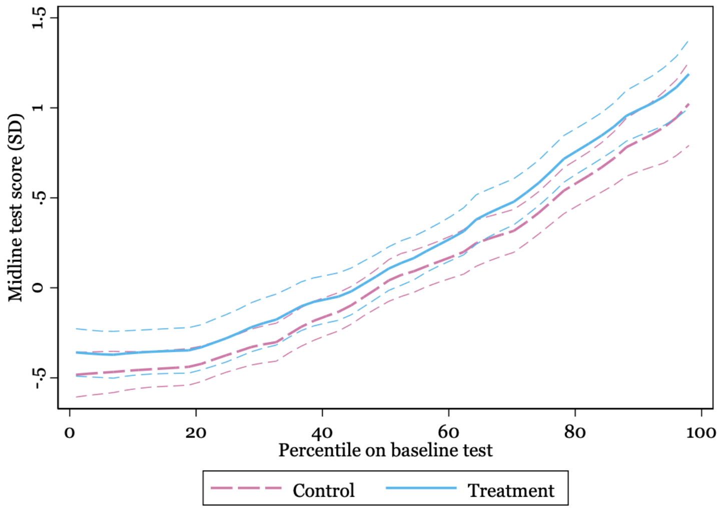
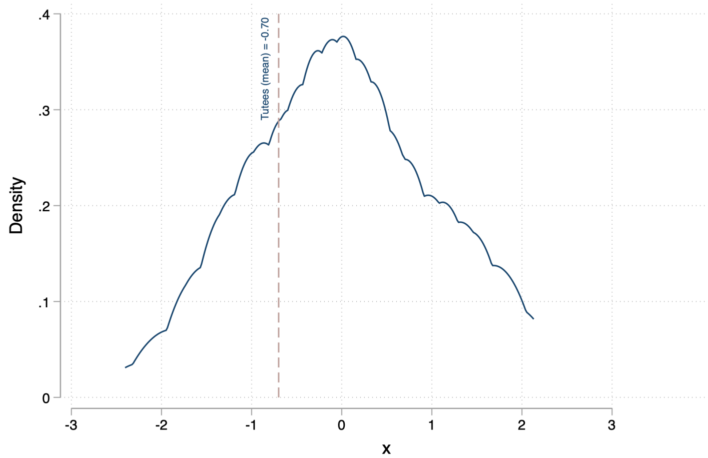


Figure 6: Non-Parametric Estimates of Treatment Effects by Baseline Performance



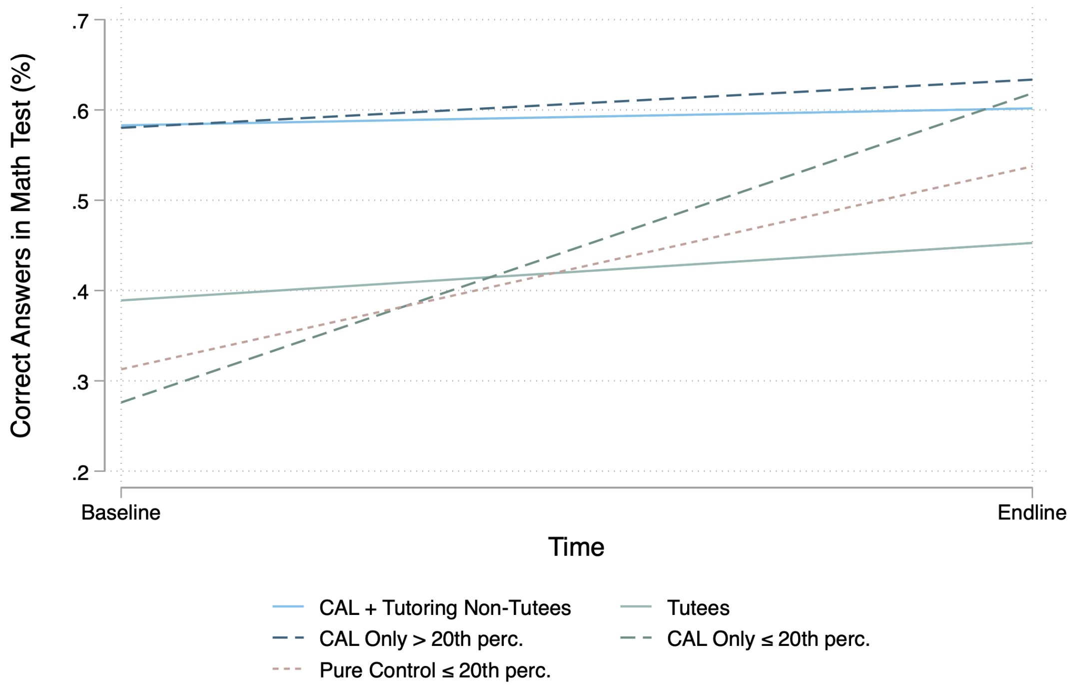
Notes: Following Muralidharan et al. (2019), the figure presents kernel-weighted local mean smoothed plots relating endline test scores to percentiles in baseline tests, separately for the treatment and control groups. The dashed lines show 95% confidence intervals.

Figure 7: Selection of Tutees and Distribution of Performance



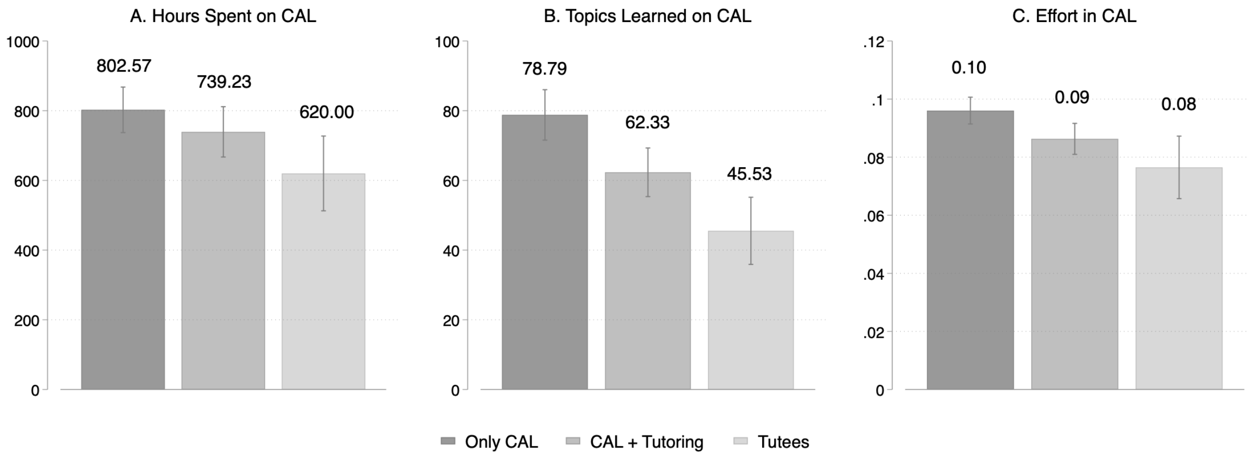
Notes: The blue line shows the average baseline score for selected tutees.

Figure 8: Share of Correct Answers by Performance Groups



Notes: The 20th percentile or below threshold of performance is used as a counterfactual for tutees. "CAL+Tutoring Non-Tutees" shows trends between baseline and endline for students who did not directly receive the group tutoring intervention but were in classrooms where approx. 20 percent of students did. "Tutees" are students who were assigned to the tutoring intervention in selected classrooms. "CAL-Only $>$ 20th perc." are students in CAL-only classrooms who are above the 20th percentile of performance. "CAL Only $\leq$ 20th percentile" are students in CAL-only classrooms who were below the 20th percentile of performance. "Pure Control $\leq$ 20th percentile" are students below the 20th percentile performance threshold in control classrooms. Dashed lines are approximate counterfactual trends for students in group tutoring classrooms who did not receive tutoring (blue) and students in tutoring classrooms who did receive tutoring (green and pink).

Figure 9: Platform Use by Groups



Notes: The CAL+Tutoring group excludes tutees.

Table 1: Balance Table

	Control		Treatment			
Variable	N	Mean/(SE)	N	Mean/(SE)	N	Difference
Female	669	0.552	664	0.566	1333	-0.015
		(0.019)		(0.019)		
Age	615	14.525	629	14.458	1244	0.067
		(0.033)		(0.033)		
Math test scores	620	6.281	631	6.442	1251	-0.162
		(0.108)		(0.110)		
Mother went to college	622	0.293	636	0.319	1258	-0.027
		(0.018)		(0.018)		
Father went to college	622	0.186	636	0.212	1258	-0.026
		(0.016)		(0.016)		
Mother went to high school	622	0.693	636	0.708	1258	-0.015
		(0.019)		(0.018)		
Father went to high school	622	0.564	636	0.558	1258	0.006
		(0.020)		(0.020)		
F-test of joint significance						0.749
F-test, number of observations						1237

Notes: *** p<0.01, ** p<0.05, * p<0.1. Math test scores are based on a short assessment administered at baseline.

Table 2: Attrition between Baseline, Midline, and Endline

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Midline Survey				Endline Survey			
	All	All	Treat	Control	All	All	Treat	Control
Treatment	0.005	0.017			0.029*	0.032*		
	(0.015)	(0.015)			(0.015)	(0.016)		
Student's age		-0.011	-0.024*	0.004		-0.029**	-0.037**	-0.022
		(0.011)	(0.014)	(0.018)		(0.012)	(0.015)	(0.020)
Student is female		-0.018	-0.006	-0.025		0.008	0.039*	-0.018
		(0.016)	(0.021)	(0.024)		(0.017)	(0.022)	(0.027)
Father is unemployed		-0.027	-0.036	-0.006		-0.009	0.030	-0.035
		(0.027)	(0.037)	(0.041)		(0.030)	(0.039)	(0.047)
Mother is unemployed		-0.024	0.005	-0.050**		-0.002	0.011	-0.007
		(0.017)	(0.023)	(0.025)		(0.019)	(0.024)	(0.029)
Baseline test score		0.001	0.000	0.004		0.006*	0.009**	0.004
		(0.003)	(0.004)	(0.005)		(0.003)	(0.004)	(0.005)
Repeated a school year		0.002	0.026	-0.023		0.004	0.035	-0.023
		(0.025)	(0.032)	(0.039)		(0.027)	(0.034)	(0.044)
Mother went to college		0.025	0.036	0.019		-0.015	-0.024	-0.004
		(0.018)	(0.024)	(0.028)		(0.020)	(0.025)	(0.032)
Mother went to high school		0.032*	0.011	0.045		0.024	0.012	0.034
		(0.019)	(0.026)	(0.029)		(0.021)	(0.027)	(0.033)
Father went to high school		-0.037**	0.007	-0.076***		-0.032*	-0.044*	-0.018
		(0.018)	(0.024)	(0.026)		(0.019)	(0.025)	(0.030)
Father went to college		-0.012	-0.016	-0.017		-0.007	0.008	-0.022
		(0.020)	(0.026)	(0.033)		(0.022)	(0.027)	(0.037)
Locus of control		0.068	0.103	0.028		0.008	-0.002	0.011
		(0.063)	(0.085)	(0.095)		(0.070)	(0.091)	(0.109)
Self-efficacy		-0.061	-0.099	-0.028		-0.068	-0.096	-0.057
		(0.049)	(0.066)	(0.073)		(0.054)	(0.071)	(0.083)
Spanish grade in 2022/23		0.002**	0.003*	0.002		-0.001	-0.001	-0.001
		(0.001)	(0.001)	(0.002)		(0.001)	(0.002)	(0.002)
Math grade in 2022/23		-0.002	-0.003	-0.002		0.003**	0.002	0.003
		(0.001)	(0.002)	(0.002)		(0.001)	(0.002)	(0.002)
Constant	0.916***	1.096***	1.286***	0.927***	0.900***	1.242***	1.327***	1.204***
	(0.011)	(0.188)	(0.251)	(0.290)	(0.011)	(0.208)	(0.266)	(0.331)
R-squared	0.000	0.037	0.043	0.055	0.003	0.033	0.051	0.033
N	1333	948	473	475	1333	948	473	475

Notes: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Standard errors in parentheses.

Table 3: Treatment Effects on Math Assessments

	Midline Math Test		Endline Math Test	
	(1)	(2)	(3)	(4)
CAL Treatment (β_1)	0.206	0.167	0.313*	0.285**
	[0.179]	[0.180]	[0.057]	[0.033]
CAL x Tutoring Treatment (β_2)	-0.103	-0.066	-0.239	-0.194
	[0.619]	[0.656]	[0.214]	[0.208]
CAL + Tutoring Treatment (β_1 + β_2)	0.103	0.101	0.073	0.091
	[0.485]	[0.401]	[0.653]	[0.518]
Control Mean	0.00	0.00	0.00	0.00
Adjusted R^2	0.01	0.28	0.02	0.30
Observations	1225	1225	1218	1218
Covariates	No	Yes	No	Yes

Notes: *** p<0.01, ** p<0.05, * p<0.1. Covariates include students' gender, age, baseline math test scores, proxies for confidence and preferences towards math, self-reported grades for the previous year and parents' employment status. When available, we also control for the baseline level of the outcome variable. The p-values shown in brackets represent the probability of obtaining an estimated treatment effect at least as large in absolute value as the one observed in our experiment, under the sharp null hypothesis that the treatment effect is zero for all observations. Randomization inference is based on 2,000 random draws.

Table 4: Treatment Effects on Math Intrinsic Motivation

	Math Intrinsic Motivation Index	I enjoy learning math	I wish I did not have to study math	I like to solve math problems
	(1)	(2)	(3)	(4)
CAL Treatment (β_1)	-0.026	0.036	-0.002	-0.133**
	[0.594]	[0.732]	[0.841]	[0.022]
CAL x Tutoring Treatment (β_2)	0.102	0.071	-0.009	0.093
	[0.349]	[0.480]	[0.891]	[0.310]
CAL + Tutoring Treatment (β_1 + β_2)	0.076	0.106	-0.011	-0.041
	[0.578]	[0.223]	[0.987]	[0.395]
Adjusted R^2	-0.00	-0.00	-0.00	0.00
Observations	1194	1191	1179	1181

Notes: *** p<0.01, ** p<0.05, * p<0.1. Covariates include students' gender, age, baseline math test scores, proxies for confidence and preferences towards math, self-reported grades for the previous year and parents' employment status. When available, we also control for the baseline level of the outcome variable. The p-values shown in brackets represent the probability of obtaining an estimated treatment effect at least as large in absolute value as the one observed in our experiment, under the sharp null hypothesis that the treatment effect is zero for all observations. Randomization inference is based on 2,000 random draws.

Table 5: Treatment Effects on Math Preferences

	Student Likes Math			
	Midline Preferences		Endline Preferences	
	(1)	(2)	(3)	(4)
CAL Treatment (β_1)	-0.051	0.005	-0.021	0.020
	[0.560]	[0.944]	[0.832]	[0.809]
CAL x Tutoring Treatment (β_2)	-0.030	-0.038	0.023	0.024
	[0.819]	[0.701]	[0.852]	[0.835]
CAL + Tutoring Treatment (β_1 + β_2)	-0.081	-0.032	0.002	0.044
	[0.379]	[0.669]	[0.984]	[0.618]
Control Mean	0.00	0.00	0.00	0.00
Adjusted R^2	-0.00	0.32	-0.00	0.24
Observations	1218	1218	1210	1210
Covariates	No	Yes	No	Yes

Notes: *** p<0.01, ** p<0.05, * p<0.1. Covariates include students' gender, age, baseline math test scores, proxies for confidence and preferences towards math, self-reported grades for the previous year and parents' employment status. When available, we also control for the baseline level of the outcome variable. The p-values shown in brackets represent the probability of obtaining an estimated treatment effect at least as large in absolute value as the one observed in our experiment, under the sharp null hypothesis that the treatment effect is zero for all observations. Randomization inference is based on 2,000 random draws.

Table 6: Treatment Effects on Absence Rates

	Share of Days Absent	
	(1)	(2)
CAL Treatment (β_1)	-0.007	-0.002
	[0.697]	[0.697]
CAL x Tutoring Treatment (β_2)	0.010	0.010
	[0.535]	[0.535]
CAL + Tutoring Treatment (β_1 + β_2)	0.008	0.008
	[0.658]	[0.658]
Control Mean	0.09	0.09
Adjusted R^2	0.00	0.00
Observations	1254	1254
Covariates	No	Yes

Notes: *** p<0.01, ** p<0.05, * p<0.1. Covariates include students' gender, age, baseline math test scores, proxies for confidence and preferences towards math, self-reported grades for the previous year and parents' employment status. When available, we also control for the baseline level of the outcome variable. The p-values shown in brackets represent the probability of obtaining an estimated treatment effect at least as large in absolute value as the one observed in our experiment, under the sharp null hypothesis that the treatment effect is zero for all observations. Randomization inference is based on 2,000 random draws. Attendance measures are specific to math classes.

Table 7: Treatment Effects on School Dropout

	School Dropout	
	(1)	(2)
CAL Treatment (β_1)	-0.019	-0.010
	[0.442]	[0.603]
CAL x Tutoring Treatment (β_2)	-0.009	-0.016
	[0.806]	[0.564]
CAL + Tutoring Treatment (β_1 + β_2)	-0.028	-0.026
	[0.263]	[0.170]
Control Mean	0.08	0.08
Adjusted R^2	0.00	0.06
Observations	1333	1333
Covariates	No	Yes

Notes: *** p<0.01, ** p<0.05, * p<0.1. Covariates include students' gender, age, baseline math test scores, proxies for confidence and preferences towards math, self-reported grades for the previous year and parents' employment status. When available, we also control for the baseline level of the outcome variable. The p-values shown in brackets represent the probability of obtaining an estimated treatment effect at least as large in absolute value as the one observed in our experiment, under the sharp null hypothesis that the treatment effect is zero for all observations. Randomization inference is based on 2,000 random draws.

Table 8: Treatment Effects on School Grades

	Math Grades		Spanish Grades	
	(1)	(2)	(3)	(4)
Panel A: Unconditional School Grades				
CAL Treatment (β_1)	-0.141	-0.153	0.012	-0.015
	[0.311]	[0.262]	[0.909]	[0.890]
CAL x Tutoring Treatment (β_2)	-0.095	-0.050	-0.072	-0.012
	[0.597]	[0.743]	[0.722]	[0.929]
CAL + Tutoring Treatment (β_1 + β_2)	-0.235	-0.203	-0.060	-0.027
	[0.102]	[0.146]	[0.599]	[0.779]
Adjusted R^2	0.01	0.31	-0.00	0.32
Observations	1247	1247	1247	1247
Covariates	No	Yes	No	Yes
Panel B: Conditional School Grades				
CAL Treatment (β_1)	-0.188	-0.202	0.029	0.006
	[0.227]	[0.158]	[0.812]	[0.959]
CAL x Tutoring Treatment (β_2)	-0.126	-0.069	-0.063	-0.009
	[0.520]	[0.666]	[0.772]	[0.953]
CAL + Tutoring Treatment (β_1 + β_2)	-0.313	-0.271	-0.034	-0.003
	[0.050]	[0.066]	[0.792]	[0.980]
Adjusted R^2	0.02	0.37	-0.00	0.31
Observations	1247	1247	1247	1247
Covariates	No	Yes	No	Yes

Notes: *** p<0.01, ** p<0.05, * p<0.1. Panel A reports results for unconditional school grades, calculated as the average of all quarterly grades available for the 2023-2024 school year. In contrast, Panel B shows conditional school grades, which reflect the final grade reported at the end of the school year. Unlike unconditional grades, conditional grades are only available for students who remained enrolled through the end of the academic year. Covariates include students' gender, age, baseline math test scores, and proxies for confidence and preferences towards math, along with self-reported grades for the previous year and parents' employment status. When available, we also control for the baseline level of the outcome variable. The p-values shown in brackets represent the probability of obtaining an estimated treatment effect at least as large in absolute value as the one observed in our experiment, under the sharp null hypothesis that the treatment effect is zero for all observations. Randomization inference is based on 2,000 random draws.

Table 9: Treatment Effects on Math Self-Beliefs and Self-Confidence

	Midline Math Self-Beliefs		Endline Math Self-Beliefs	
	(1)	(2)	(3)	(4)
Panel A: Math Self-Beliefs (SD)				
CAL Treatment (β_1)	0.117	0.127	0.119	0.134
	[0.237]	[0.186]	[0.156]	[0.100]
CAL x Tutoring Treatment (β_2)	-0.009	0.020	0.007	0.035
	[0.941]	[0.872]	[0.938]	[0.725]
CAL + Tutoring Treatment (β_1 + β_2)	0.108	0.147	0.126	0.169**
	[0.254]	[0.111]	[0.133]	[0.038]
Adjusted R^2	0.00	0.15	0.00	0.14
Observations	1205	1205	1188	1188
Covariates	No	Yes	No	Yes
Panel B: Math Self-Confidence Index				
CAL Treatment (β_1)	-0.016	-0.007	-0.010	-0.002
	[0.274]	[0.605]	[0.603]	[0.923]
CAL x Tutoring Treatment (β_2)	0.011	0.014	-0.005	-0.000
	[0.599]	[0.368]	[0.820]	[0.999]
CAL + Tutoring Treatment (β_1 + β_2)	-0.005	0.008	-0.014	-0.002
	[0.713]	[0.533]	[0.422]	[0.918]
Control Mean	0.66	0.66	0.64	0.64
Adjusted R^2	0.00	0.34	-0.00	0.28
Observations	1225	1225	1206	1206
Covariates	No	Yes	No	Yes

Notes: *** p<0.01, ** p<0.05, * p<0.1. Covariates include students' gender, age, baseline math test scores, proxies for confidence and preferences towards math, self-reported grades for the previous year and parents' employment status. When available, we also control for the baseline level of the outcome variable. The p-values shown in brackets represent the probability of obtaining an estimated treatment effect at least as large in absolute value as the one observed in our experiment, under the sharp null hypothesis that the treatment effect is zero for all observations. Randomization inference is based on 2,000 random draws. The outcome variable in Panel A is a standardized measure of students' self-reported expected performance on a math test. The outcome variable in Panel B is a standardized index based on students' answers to the following items: 1) When teachers teach me a new math topic, I learn it without much difficulty; 2) When I take a math test, I am confident that I can answer the questions well; 3) I can help my classmates with math homework that they don't understand; 4) I feel more capable as I learn more things in math. Answers ranged 1 (Never) to 4 (Always).

Table 10: Treatment Effects on Math Self-Concept

	Math Self-Concept Index	I usually do well in math	Math is harder for me than for many of my classmates
	(1)	(2)	(3)
CAL Treatment (β_1)	-0.008	-0.005	0.032
	[0.844]	[0.947]	[0.486]
CAL x Tutoring Treatment (β_2)	0.011	-0.003	-0.093
	[0.927]	[0.846]	[0.292]
CAL + Tutoring Treatment (β_1 + β_2)	0.003	-0.008	-0.060
	[0.832]	[0.648]	[0.527]
Adjusted R^2	-0.00	-0.00	-0.00
Observations	1184	1182	1175

Notes: *** p<0.01, ** p<0.05, * p<0.1. Covariates include students' gender, age, baseline math test scores, proxies for confidence and preferences towards math, self-reported grades for the previous year and parents' employment status. When available, we also control for the baseline level of the outcome variable. The p-values shown in brackets represent the probability of obtaining an estimated treatment effect at least as large in absolute value as the one observed in our experiment, under the sharp null hypothesis that the treatment effect is zero for all observations. Randomization inference is based on 2,000 random draws.

Table 11: Treatment Effects on Math Overconfidence

	Midline Overconfidence		Endline Overconfidence	
	(1)	(2)	(3)	(4)
CAL Treatment (β_1)	0.018	0.032	-0.002	0.008
	[0.698]	[0.499]	[0.956]	[0.844]
CAL x Tutoring Treatment (β_2)	0.026	0.029	0.042	0.050
	[0.645]	[0.609]	[0.331]	[0.269]
CAL + Tutoring Treatment (β_1 + β_2)	0.044	0.061	0.040	0.058
	[0.351]	[0.203]	[0.289]	[0.132]
Adjusted R^2	-0.00	0.05	-0.00	0.05
Observations	1205	1205	1188	1188
Covariates	No	Yes	No	Yes

Notes: *** p<0.01, ** p<0.05, * p<0.1. Notes: *** p<0.01, ** p<0.05, * p<0.1. The outcome variable takes the value of 1 if the teacher switched classes after program assignment, and 0 otherwise. Covariates include students' gender, age, baseline math test scores, proxies for confidence and preferences towards math, self-reported grades for the previous year and parents' employment status. When available, we also control for the baseline level of the outcome variable. The p-values shown in brackets represent the probability of obtaining an estimated treatment effect at least as large in absolute value as the one observed in our experiment, under the sharp null hypothesis that the treatment effect is zero for all observations. Randomization inference is based on 2,000 random draws.

Table 12: Treatment Effects on Self-Efficacy Index

	Midline Self-Efficacy Index		Endline Self-Efficacy Index	
	(1)	(2)	(3)	(4)
CAL Treatment (β_1)	-0.016	-0.018	-0.009	-0.012
	[0.284]	[0.151]	[0.534]	[0.361]
CAL x Tutoring Treatment (β_2)	0.017	0.020	-0.000	0.006
	[0.368]	[0.126]	[0.986]	[0.737]
CAL + Tutoring Treatment (β_1 + β_2)	0.001	0.002	-0.010	-0.006
	[0.941]	[0.875]	[0.507]	[0.625]
Adjusted R^2	0.00	0.25	-0.00	0.13
Observations	1217	1217	1194	1194
Covariates	No	Yes	No	Yes

Notes: *** p<0.01, ** p<0.05, * p<0.1. Notes: *** p<0.01, ** p<0.05, * p<0.1. Covariates include students' gender, age, baseline math test scores, proxies for confidence and preferences towards math, self-reported grades for the previous year and parents' employment status. When available, we also control for the baseline level of the outcome variable. The p-values shown in brackets represent the probability of obtaining an estimated treatment effect at least as large in absolute value as the one observed in our experiment, under the sharp null hypothesis that the treatment effect is zero for all observations. Randomization inference is based on 2,000 random draws.

Table 13: Treatment Effects on Grit Index

	Grit Index	
	(1)	(2)
CAL Treatment (β_1)	0.011	0.006
	[0.430]	[0.676]
CAL x Tutoring Treatment (β_2)	0.001	0.001
	[0.965]	[0.968]
CAL + Tutoring Treatment (β_1 + β_2)	0.012	0.006
	[0.361]	[0.653]
Adjusted R^2	-0.00	0.03
Observations	1333	1333
Covariates	No	Yes

Notes: *** p<0.01, ** p<0.05, * p<0.1. Covariates include students' gender, age, baseline math test scores, proxies for confidence and preferences towards math, self-reported grades for the previous year and parents' employment status. When available, we also control for the baseline level of the outcome variable. The p-values shown in brackets represent the probability of obtaining an estimated treatment effect at least as large in absolute value as the one observed in our experiment, under the sharp null hypothesis that the treatment effect is zero for all observations. Randomization inference is based on 2,000 random draws. The outcome is a standardized variable based on students' answers to the following items: 1) The schoolwork I like the most is the one that makes me think, 2) If I think I will lose in a game, I do not want to continue playing, 3) If I set a goal and see that it's harder than I though I lose interest, and 4) I work hard on tasks. Answer options ranged from 1 (Doesn't seem like me at all) to 5 (Very much like me).

Table 14: Treatment Ettects on Logic Task

	Logic Task: Easy	Logic Task: difficult	Logic Task: Give Up
	(1)	(2)	(3)
CAL Treatment (β_1)	0.087	-0.094	0.007
	[0.075]	[0.019]	[0.730]
CAL x Tutoring Treatment (β_2)	-0.037	0.023	0.014
	[0.651]	[0.851]	[0.720]
CAL + Tutoring Treatment (β_1 + β_2)	0.050	-0.071	0.021
	[0.201]	[0.043]	[0.417]
Control Mean	0.43	0.38	0.19
Adjusted R^2	0.00	0.07	-0.00
Observations	1211	1211	1211

Notes: *** p<0.01, ** p<0.05, * p<0.1. Covariates include students' gender, age, baseline math test scores, proxies for confidence and preferences towards math, self-reported grades for the previous year and parents' employment status. When available, we also control for the baseline level of the outcome variable. The p-values shown in brackets represent the probability of obtaining an estimated treatment ettect at least as large in absolute value as the one observed in our experiment, under the sharp null hypothesis that the treatment ettect is zero for all observations. Randomization inference is based on 2,000 random draws.

Table 15: Treatment Effects on Locus of Control

	Locus of Control	
	(1)	(2)
CAL Treatment (β_1)	0.010	0.004
	[0.258]	[0.585]
CAL x Tutoring Treatment (β_2)	-0.025*	-0.020*
	[0.056]	[0.056]
CAL + Tutoring Treatment (β_1 + β_2)	-0.014	-0.015*
	[0.120]	[0.056]
Control Mean	0.67	0.67
Adjusted R^2	0.00	0.11
Observations	1196	1196
Covariates	No	Yes

Notes: *** p<0.01, ** p<0.05, * p<0.1. Covariates include students' gender, age, baseline math test scores, proxies for confidence and preferences towards math, self-reported grades for the previous year and parents' employment status. When available, we also control for the baseline level of the outcome variable. The p-values shown in brackets represent the probability of obtaining an estimated treatment effect at least as large in absolute value as the one observed in our experiment, under the sharp null hypothesis that the treatment effect is zero for all observations. Randomization inference is based on 2,000 random draws.

Table 16: Treatment Effects on Student Time Use

	Hours spent on:			
	Academic Activities	Work	Entertainment	Sleep
	(1)	(2)	(3)	(4)
CAL Treatment (β_1)	0.150	-0.163	0.159	0.001
	[0.607]	[0.773]	[0.499]	[0.956]
CAL x Tutoring Treatment (β_2)	-0.191	-0.375	-0.223	0.105
	[0.503]	[0.697]	[0.395]	[0.556]
CAL + Tutoring Treatment (β_1 + β_2)	-0.041	-0.538	-0.223	0.106
	[0.838]	[0.277]	[0.539]	[0.385]
Control Mean	7.90	8.61	7.55	5.66
Adjusted R^2	-0.00	-0.00	-0.00	-0.00
Observations	1194	1182	1185	1115

Notes: *** p<0.01, ** p<0.05, * p<0.1. Covariates include students' gender, age, baseline math test scores, proxies for confidence and preferences towards math, self-reported grades for the previous year and parents' employment status. When available, we also control for the baseline level of the outcome variable. The p-values shown in brackets represent the probability of obtaining an estimated treatment effect at least as large in absolute value as the one observed in our experiment, under the sharp null hypothesis that the treatment effect is zero for all observations. Randomization inference is based on 2,000 random draws.

Table 17: Treatment Effects on Student Depression

	Depression Index	
	(1)	(2)
CAL Treatment (β_1)	0.012	0.009
	[0.286]	[0.306]
CAL x Tutoring Treatment (β_2)	-0.003	-0.003
	[0.847]	[0.755]
CAL + Tutoring Treatment (β_1 + β_2)	0.009	0.006
	[0.396]	[0.511]
Adjusted R^2	-0.00	0.08
Observations	1205	1205
Covariates	No	Yes

Notes: *** p<0.01, ** p<0.05, * p<0.1. Covariates include students' gender, age, baseline math test scores, proxies for confidence and preferences towards math, self-reported grades for the previous year and parents' employment status. When available, we also control for the baseline level of the outcome variable. The p-values shown in brackets represent the probability of obtaining an estimated treatment effect at least as large in absolute value as the one observed in our experiment, under the sharp null hypothesis that the treatment effect is zero for all observations. Randomization inference is based on 2,000 random draws. The outcome variable is a standardized index based on students' answers to the following items: 1) I am happy, 2) I worry a lot, 3) I feel sad, 4) I get upset quickly, 5) I often think I did something wrong, 6) It's hard for me to concentrate, 7) I feel lonely, and 8) I am not happy. Answers range from 1 (Strongly disagree) to 4 (Strongly agree).

Table 18: Treatment Effects on Math Anxiety

	Math Anxiety Index	I'm worried of having bad grades in math	I get nervous before math tests
	(1)	(2)	(3)
CAL Treatment (β_1)	0.135	0.033	0.060
	[0.078]	[0.265]	[0.049]
CAL x Tutoring Treatment (β_2)	-0.123	-0.016	-0.061
	[0.229]	[0.657]	[0.115]
CAL + Tutoring Treatment (β_1 + β_2)	0.012	0.017	-0.001
	[0.229]	[0.598]	[0.819]
Control Mean	0.00	0.77	0.71
Adjusted R^2	0.00	0.00	0.00
Observations	1182	1199	1183

Notes: *** p<0.01, ** p<0.05, * p<0.1. Covariates include students' gender, age, baseline math test scores, proxies for confidence and preferences towards math, self-reported grades for the previous year and parents' employment status. When available, we also control for the baseline level of the outcome variable. The p-values shown in brackets represent the probability of obtaining an estimated treatment effect at least as large in absolute value as the one observed in our experiment, under the sharp null hypothesis that the treatment effect is zero for all observations. Randomization inference is based on 2,000 random draws.

Table 19: Treatment Effects on Teacher Behavior

	Teacher Behavior	
	(1)	(2)
Panel A: Time Mismanagement Index		
CAL Treatment (β_1)	-0.082	-0.030
	[0.556]	[0.801]
CAL x Tutoring Treatment (β_2)	-0.145	-0.146
	[0.382]	[0.301]
CAL + Tutoring Treatment (β_1 + β_2)	-0.145	-0.146
	[0.382]	[0.301]
Adjusted R^2	0.01	0.12
Observations	1148	1148
Covariates	No	Yes
Panel B: Teacher Engagement Index		
CAL Treatment (β_1)	-0.181	-0.144
	[0.102]	[0.112]
CAL x Tutoring Treatment (β_2)	0.044	0.041
	[0.773]	[0.763]
CAL + Tutoring Treatment (β_1 + β_2)	-0.137	-0.103
	[0.220]	[0.249]
Adjusted R^2	0.00	0.12
Observations	1151	1151
Covariates	No	Yes

Notes: *** p<0.01, ** p<0.05, * p<0.1. Covariates include students' gender, age, baseline math test scores, proxies for confidence and preferences towards math, self-reported grades for the previous year and parents' employment status. The p-values shown in brackets represent the probability of obtaining an estimated treatment effect at least as large in absolute value as the one observed in our experiment, under the sharp null hypothesis that the treatment effect is zero for all observations. Randomization inference is based on 2,000 random draws.

Table 20: Treatment Effects on Teaching Effectiveness Index

	Teacher Effectiveness Index		Teacher Ineffectiveness Index	
	(1)	(2)	(3)	(4)
CAL Treatment (β_1)	0.055	0.059	0.068	0.083
	[0.606]	[0.579]	[0.480]	[0.390]
CAL x Tutoring Treatment (β_2)	-0.144	-0.122	-0.155	-0.148
	[0.294]	[0.359]	[0.245]	[0.245]
CAL + Tutoring Treatment (β_1 + β_2)	-0.089	-0.063	-0.087	-0.066
	[0.403]	[0.555]	[0.379]	[0.530]
Adjusted R^2	0.00	0.04	0.00	0.03
Observations	1161	1161	1165	1165
Covariates	No	Yes	No	Yes

Notes: *** p<0.01, ** p<0.05, * p<0.1. Covariates include students' gender, age, baseline math test scores, proxies for confidence and preferences towards math, self-reported grades for the previous year and parents' employment status. The p-values shown in brackets represent the probability of obtaining an estimated treatment effect at least as large in absolute value as the one observed in our experiment, under the sharp null hypothesis that the treatment effect is zero for all observations. Randomization inference is based on 2,000 random draws

Table 21: Treatment Effects on Test Scores by Content Level

	Below Grade-Level		At Grade-Level	
	(1)	(2)	(3)	(4)
CAL Treatment (β_1)	0.282**	0.262**	0.243	0.214*
	[0.034]	[0.028]	[0.102]	[0.073]
CAL x Tutoring Treatment (β_2)	-0.163	-0.127	-0.262	-0.220
	[0.304]	[0.379]	[0.153]	[0.118]
CAL + Tutoring Treatment (β_1 + β_2)	0.119	0.135	-0.019	-0.006
	[0.400]	[0.340]	[0.901]	[0.955]
Adjusted R^2	0.01	0.22	0.01	0.24
Observations	1218	1218	1217	1217
Covariates	No	Yes	No	Yes

Notes: *** p<0.01, ** p<0.05, * p<0.1. Covariates include students' gender, age, baseline math test scores, proxies for confidence and preferences towards math, self-reported grades for the previous year and parents' employment status. When available, we also control for the baseline level of the outcome variable. The p-values shown in brackets represent the probability of obtaining an estimated treatment effect at least as large in absolute value as the one observed in our experiment, under the sharp null hypothesis that the treatment effect is zero for all observations. Randomization inference is based on 2,000 random draws.

Table 22: Treatment Effects on Test Scores by Math Area

	Midline			Endline			
	Algebra	Numbers	Data and Prob	Algebra	Numbers	Data and Prob	Geometry
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
CAL Treatment (β_1)	0.195*	0.093	-0.011	0.214*	0.317**	0.028	0.133
	[0.100]	[0.376]	[0.929]	[0.088]	[0.022]	[0.679]	[0.229]
CAL x Tutoring (β_2)	-0.091	0.011	-0.062	-0.220	-0.106	0.056	-0.191
	[0.503]	[0.910]	[0.460]	[0.147]	[0.451]	[0.693]	[0.101]
CAL x Tutoring (β_1 + β_2)	0.104	0.104	-0.073	-0.006	0.210	0.084	-0.058
	[0.380]	[0.376]	[0.380]	[0.147]	[0.149]	[0.382]	[0.490]
Adjusted R^2	0.01	0.19	-0.00	0.01	0.02	-0.00	0.00
Observations	1225	1225	1220	1217	1217	1218	1218

Notes: *** p<0.01, ** p<0.05, * p<0.1. Covariates include students' gender, age, baseline math test scores, proxies for confidence and preferences towards math, self-reported grades for the previous year and parents' employment status. When available, we also control for the baseline level of the outcome variable. The p-values shown in brackets represent the probability of obtaining an estimated treatment effect at least as large in absolute value as the one observed in our experiment, under the sharp null hypothesis that the treatment effect is zero for all observations. Randomization inference is based on 2,000 random draws.

Table 23: Treatment Effects on Test Scores by Knowledge Domain

	Midline			Endline		
	Knowing	Applying	Reasoning	Knowing	Applying	Reasoning
	(1)	(2)	(3)	(4)	(5)	(6)
CAL Treatment (β_1)	0.198	0.075	0.075	0.311***	0.164	0.211**
	[0.170]	[0.346]	[0.380]	[0.007]	[0.182]	[0.042]
CAL x Tutoring (β_2)	0.009	-0.064	-0.156	-0.213	-0.097	-0.089
	[0.916]	[0.547]	[0.160]	[0.120]	[0.456]	[0.372]
CAL + Tutoring ($\beta_1 + \beta_2$)	0.009	-0.064	-0.156	-0.213	-0.097	-0.089
	[0.916]	[0.547]	[0.160]	[0.120]	[0.456]	[0.372]
Adjusted R^2	0.01	0.18	0.00	0.02	0.00	0.01
Observations	1225	1225	1224	1218	1217	1218

Notes: *** p<0.01, ** p<0.05, * p<0.1. Covariates include students' gender, age, baseline math test scores, proxies for confidence and preferences towards math, self-reported grades for the previous year and parents' employment status. When available, we also control for the baseline level of the outcome variable. The p-values shown in brackets represent the probability of obtaining an estimated treatment effect at least as large in absolute value as the one observed in our experiment, under the sharp null hypothesis that the treatment effect is zero for all observations. Randomization inference is based on 2,000 random draws.

Table 24: Treatment Effects on Math Assessments by Levels of Locus of Control

	Midline Math Test		Endline Math Test	
	(1)	(2)	(3)	(4)
CAL Treatment, Low LOC	0.234	0.179	0.295	0.269
	[0.501]	[0.503]	[0.485]	[0.495]
CAL Treatment, High LOC	0.178	0.154	0.332**	0.303**
	[0.220]	[1.000]	[0.033]	[0.022]
CAL+ Tutoring Treatment, High LOC	0.273*	0.260**	0.200	0.204
	[0.068]	[0.033]	[0.196]	[0.122]
CAL + Tutoring Treatment, Low LOC	-0.006	-0.014	0.002	0.017
	[0.973]	[0.903]	[0.992]	[0.897]
Control Mean	0.00	0.00	0.00	0.00
Adjusted R^2	0.03	0.28	0.05	0.31
Observations	1225	1225	1218	1218
Covariates	No	Yes	No	Yes

Notes: *** p<0.01, ** p<0.05, * p<0.1. Covariates include students' gender, age, baseline math test scores, proxies for confidence and preferences towards math, self-reported grades for the previous year and parents' employment status. When available, we also control for the baseline level of the outcome variable. The p-values shown in brackets represent the probability of obtaining an estimated treatment effect at least as large in absolute value as the one observed in our experiment, under the sharp null hypothesis that the treatment effect is zero for all observations. Randomization inference is based on 2,000 random draws.

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9 Appendix

A.1 Supplemental Tables and Figures

Table A1: Treatment Effects on Math Overconfidence

	Midline Overconfidence		Endline Overconfidence	
	(1)	(2)	(3)	(4)
CAL Treatment (β_1)	0.018	0.032	-0.002	0.008
	[0.698]	[0.499]	[0.956]	[0.844]
CAL x Tutoring Treatment (β_2)	0.026	0.029	0.042	0.050
	[0.645]	[0.609]	[0.331]	[0.269]
CAL + Tutoring Treatment (β_1 + β_2)	0.044	0.061	0.040	0.058
	[0.351]	[0.203]	[0.289]	[0.132]
Adjusted R^2	-0.00	0.05	-0.00	0.05
Observations	1205	1205	1188	1188
Covariates	No	Yes	No	Yes

Notes: *** p<0.01, ** p<0.05, * p<0.1. Notes: *** p<0.01, ** p<0.05, * p<0.1. The outcome variable takes the value of 1 if the teacher switched classes after program assignment, and 0 otherwise. Covariates include students' gender, age, baseline math test scores, proxies for confidence and preferences towards math, self-reported grades for the previous year and parents' employment status. When available, we also control for the baseline level of the outcome variable. The p-values shown in brackets represent the probability of obtaining an estimated treatment effect at least as large in absolute value as the one observed in our experiment, under the sharp null hypothesis that the treatment effect is zero for all observations. Randomization inference is based on 2,000 random draws.

Table A2: Treatment Effects on Student Time Use

	In person classes	Unpaid work (at home)	Unpaid work (out of home)	Paid work (out of school)	Study	Meeting friends	Entertainment	Sports
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
CAL Treatment	-0.060	-0.368	-0.105	-0.132	0.069	-0.072	0.007	0.008
	[0.805]	[0.057]	[0.616]	[0.617]	[0.680]	[0.590]	[0.946]	[0.926]
CAL x Tutoring	-0.160	0.153	0.018	-0.185	0.071	-0.078	0.041	0.071
	[0.584]	[0.517]	[0.952]	[0.529]	[0.677]	[0.679]	[0.749]	[0.541]
Control Mean	5.99	3.79	2.26	2.74	2.03	2.25	3.33	2.09
Adjusted R^2	0.09	0.08	0.13	0.14	0.02	0.09	0.05	0.22
Observations	1108	1096	1072	1069	1105	1091	1097	1093

Notes: *** p<0.01, ** p<0.05, * p<0.1. Covariates include students' gender, age, baseline math test scores, proxies for confidence and preferences towards math, self-reported grades for the previous year and parents' employment status. When available, we also control for the baseline level of the outcome variable. The *p*-values shown in brackets represent the probability of obtaining an estimated treatment effect at least as large in absolute value as the one observed in our experiment, under the sharp null hypothesis that the treatment effect is zero for all observations. Randomization inference is based on 2,000 random draws.

Table A3: Treatment Effects on Gender Beliefs

	Midline Gender Beliefs Index		Endline Gender Beliefs Index	
	(1)	(2)	(3)	(4)
CAL Treatment	0.093	0.055	0.037	0.040
	[0.386]	[0.540]	[0.632]	[0.621]
CAL x Tutoring Treatment	-0.107	-0.094	-0.006	-0.027
	[0.415]	[0.344]	[0.950]	[0.796]
Adjusted R^2	0.00	0.13	-0.00	0.02
Observations	1204	1204	1181	1181
Covariates	No	Yes	No	Yes

Notes: *** p<0.01, ** p<0.05, * p<0.1. Notes: *** p<0.01, ** p<0.05, * p<0.1. Covariates include students' gender, age, baseline math test scores, proxies for confidence and preferences towards math, self-reported grades for the previous year and parents' employment status. When available, we also control for the baseline level of the outcome variable. The p-values shown in brackets represent the probability of obtaining an estimated treatment effect at least as large in absolute value as the one observed in our experiment, under the sharp null hypothesis that the treatment effect is zero for all observations. Randomization inference is based on 2,000 random draws.

Table A4: Treatment Effects on Midline Math Performance (Lee Bounds)

	Midline Math Test		Endline Math Test	
	Lower Bound	Upper Bound	Lower Bound	Upper Bound
	(1)	(2)	(3)	(4)
CAL Treatment	0.317**	0.086	0.373**	0.233*
	[0.043]	[0.552]	[0.021]	[0.094]
CAL x Tutoring Treatment	-0.175	-0.011	-0.246	-0.241
	[0.395]	[0.961]	[0.223]	[0.168]
Observations	1206	1206	1200	1198

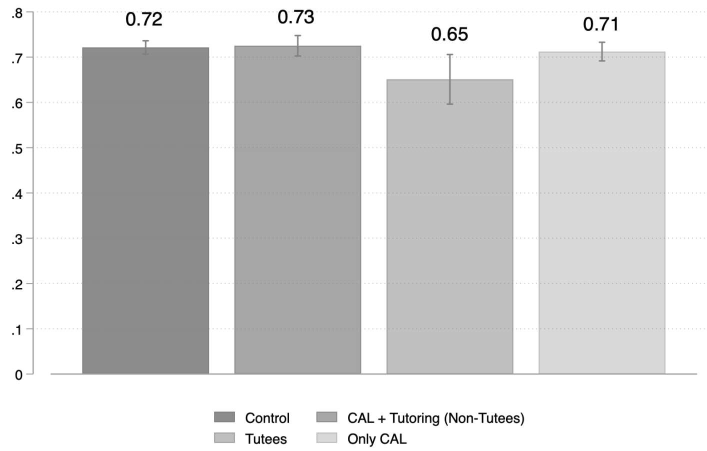
Notes: *** $p<0.01$, ** $p<0.05$, * $p<0.1$. The p-values shown in brackets represent the probability of obtaining an estimated treatment effect at least as large in absolute value as the one observed in our experiment, under the sharp null hypothesis that the treatment effect is zero for all observations. Randomization inference is based on 2,000 random draws.

Table A5: Dose-response of ALEKS Use

	Standardized Math Scores		
	IV estimates	OLS VA (full sample)	OLS VA (treatment group)
	(1)	(2)	(3)
CAL hours	0.0266***	0.0191***	0.0256***
	(0.0056)	(0.0026)	(0.0038)
Baseline math score (SD)	0.1431***	0.1456***	0.1160***
	(0.0098)	(0.0097)	(0.0137)
Observations	1218	1218	617
R-squared	0.3248	0.3296	0.3211
Difference-in-Sargan statistic for exogeneity	0.1304		
Extrapolated estimates of 2.5-hour treatment	0.0665	0.0477	0.0641
Extrapolated estimates 25 hours (1500 minutes)	0.7340	0.5267	0.6767

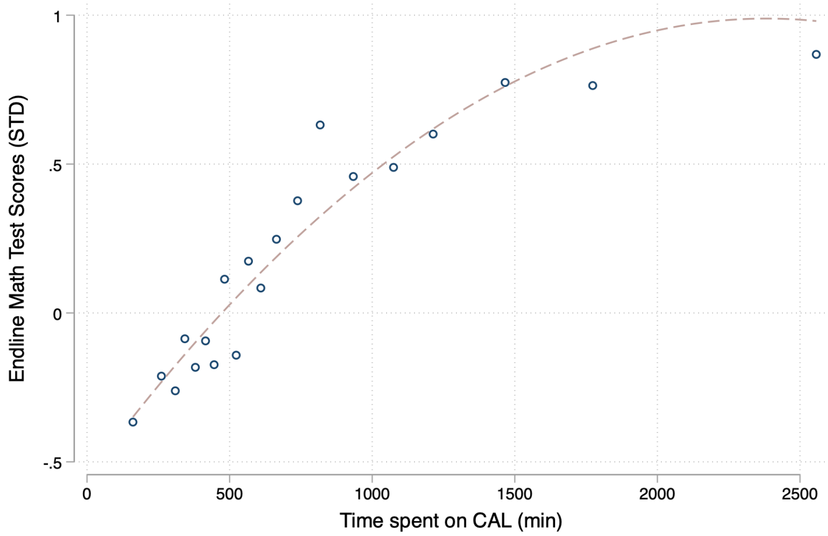
Notes: *** $p<0.01$, ** $p<0.05$, * $p<0.1$. Standard errors in parentheses. Covariates include students' gender, age, baseline math test scores, proxies for confidence and preferences towards math, self-reported grades for the previous year and parents' employment status. Column (1) instruments minutes in ALEKS with the randomized allocation. Column (2) presents OLS value-added models using the full sample. Column (3) presents OLS VA models using only data on the lottery-winners. One hour is equivalent to 60 minutes of platform use.

Figure A1: Self-Efficacy Index by Performance Groups



Notes: The CAL + Tutoring group excludes tutees.

Figure A2: Returns to Hours of CAL Use



Notes: This sample is limited to treated students (N = 664).

A.2 Additional Analysis

A.2.1 Power Calculations

To estimate statistical power, we simulate the randomization procedure described in Section 1.3 one thousand times and assume a minimum detectable effect (MDE) of 0.3 standard deviations. Evans and Yuan (2022) find this to be in the 80th percentile of effect sizes of RCTs with learning or access outcomes in low- and middle-income countries. Although this would be above the median of 0.2 for small RCTs, both tutoring and computer-adaptive learning have been identified as two of the most effective interventions to improve learning outcomes. A recent review of tutoring interventions by Nickow et al. (2020) found an overall pooled effect size estimate of 0.37 standard deviations. To the best of our knowledge, there are no meta-studies looking at effects from computer-*adaptive* learning technologies such as the one used in this study.

For this exercise, I use 2019 administrative data on 9th grade students in the same sample of schools selected for this intervention. This data includes students' test scores on a national diagnostic assessment, their sex, age, and a socioeconomic indicator. We also simulate and include a baseline test score variable. These simulations give us a statistical power of approximately 80 percent, which is the standard for randomized controlled trials in the social sciences. We aim to maximize this number by collecting prognostic variables at baseline, including tests scores on a short math assessment and other student characteristics. In Table A6, we present the results of the simulations using alternative MDEs (0.25, 0.3, 0.35).

MDE	Power
0.25	0.674
0.30	0.799
0.35	0.893

Table A6: Power Simulations